

Electrification and renewable energy nexus in developing countries; an overarching analysis of hydrogen production and electric vehicles integrality in renewable energy penetration

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ABSTRACT

As the world continues to fight against climate change, renewables have emerged as an imminent solution, however, a model that can help (developing) countries to effectively migrate to a renewable energy-based power sector is limited in literature. In this study, a comprehensive electricity generation analysis based on an hourly time-step is done in a bid to solve to electrification crisis in developing countries. This study aims to present the feasible pathways with which sustainable electrification can be achieved. This is done by considering the financial feasibility, environmental sustainability, and technical viability of different technological combinations. In comparison to existing literature, this study is novel as the integration of both renewable energy sources and a fossil fuel source is analyzed. With Nigeria being the study area, the integration of five renewable energy-based technologies namely; offshore wind power plant, onshore wind power plant, solar photovoltaic system, concentrated solar power plant, and hydropower plant as well as pumped hydro storage system is considered within the scope of this study. Based on the results of the analysis, out of the ninety-nine different combinatory scenarios modeled, the most viable solution with the use of renewables and natural gas is the combination of natural gas power plants, on-shore wind power plants, and solar photovoltaic systems. This requires a large capacity installation (32,500 MW of natural gas power plant, 12,000 MW of solar photovoltaic plant, and 7000 MW of on-shore wind power plant) as well as a total annual cost and an investment cost of \$ 22B and \$ 48.7B. Furthermore, solving the poised problem based on renewable energy technologies will only require the combination of on-shore wind power plants, hydropower plants, solar photovoltaic system, and pumped hydro storage system. The integration of plug-in battery electric vehicles and hydrogen production will help maximize the electricity produced from the renewable energy-based power systems. This will also reduce the overall carbon emission from the country at large. Beyond providing feasible scenarios and plans to tackle the existing electricity production, guidelines on steps (not to take) to tackle the electrification problem are embedded in this study.

1. Introduction

Renewable energy (RE) has been patronized more in recent decades due to its environmental sustainability and a projected scarcity of fossil fuels. However, fossil fuels still contribute about 70% of the total global

energy supply and this, in turn, increases climate change effects significantly [1]. Many countries are developing robust and comprehensive strategies to mitigate their carbon emissions and reduce greenhouse gas (GHG) emissions in general. Nevertheless, the effectiveness of these GHGs mitigating strategies towards environmental sustainability will be evaluated with time [1]. The global installed capacity of RE has been

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Nomenclature

BEV	Battery electric vehicle
CEEP	Critical excess electricity production
CHP	Combine heat and power
CSP	Concentrated solar power plant
DC	Direct Current
EV	Electric Vehicle
GDP	Gross Domestic Product
Hydro	River hydropower plant
NG	Natural gas power plant
PES	Primary energy supply

PPI	Power production importation
PumpST	Hydro-storage system pump capacity
O & M	Operation and Maintenance
RE	Renewable energy
RES	Renewable energy sources
SolarPV	Solar photovoltaic power plant
ST	Pumped hydro-storage system
TurbineST	Hydro-storage system turbine capacity
V2G	Vehicle-to-Grid
WindOFF	Offshore wind power plant
WindON	Onshore wind power plant

increasing steadily with China taking the lead with a total RE installed capacity of 789 GW (including hydropower, wind, solar, biomass). The USA and Brazil follow with a cumulative capacity of 284 GW and 142 GW, respectively [2]. Hydrogen production as a renewable energy source can play an integral role in the attainment of a secure, clean, and affordable future. Research has shown that hydrogen can solve different energy challenges [3].

Hydrogen production from RE offers a sustainable pathway to decarbonize many sectors [4]. It can also contribute to the improvement of air quality since outdoor air pollution remains a sustainable problem. Typically, outdoor air pollution leads to about 3 million premature yearly death worldwide [5]. Several fuels or energy sources such as renewable energy sources (RES), coal, natural gas, nuclear energy, and oil can be used to produce hydrogen. This is transformed into electricity using fuel cells and other technologies for domestic/industrial use [6]. Potentially, hydrogen can help solve the intermittency problem attached to RE sources like solar photovoltaics (PV) and wind systems as it can serve as a backup system [7]. In literature, the integration of fuel cells with different renewable energy systems has been studied. For instance, Sezer and Koc [8] developed and analyzed a wind, solar, and osmotic hybrid multigeneration system using a fuel cell as the backup system from a thermodynamic perspective. They concluded that the multigeneration system can work consistently for 24 h by utilizing the hydrogen produced from the excess electricity in the fuel cell [8]. Some of the challenges [6] faced in the integration of hydrogen into the global energy mix include:

- The production cost of hydrogen from low-carbon energy sources.
- The slow infrastructural development of hydrogen and limited adoption.
- Hydrogen production is almost entirely based on fossil fuels today and its production is responsible for a large volume of CO₂ emission.
- Limitations due to government policies and developments in the hydrogen industry.

The most recent advancements in wind, solar PV, electric vehicles, and batteries have proven to the world that clean energy at individual and industrial levels is feasible with adequate policies and technological innovations. Dahlke et al. [9] reviewed the policy and market drivers for advancing clean energy. They found that stimulus policies in the form of tax reliefs, feed-in-tariffs, renewable portfolio standards, as well as substantial research and development have helped overcome the commercial barriers in clean energy projects [9]. Also, an economic evaluation of government policies' contribution to affordable and clean energy showed that investment based policies are more preferable to production-based policies for microgrids [10]. In the European Union (EU), the clean energy strategy for islands' analysis reported that centralized and hierarchical spatial governance structure created an environment that argues against small and decentralized renewable energy technologies [11]. Electric vehicles (EVs) have a high potential to

reduce fossil fuel demands, and the deployment of the vehicles has been on a rapid increase in recent decades. The global fleet of commercial EVs crossed the 5 million mark in 2018 [12]. This is an increase of about 2 million (63%) compared to the previous year [13]. China hosts 45% of electric cars on the road in 2018 while Europe accounted for 24% of the global fleet, and the United States 22% [14]. By 2030, it is projected that global electric car sales would reach 23 million and the stock would exceed 130 million vehicles, this excludes two or three-wheelers. Projected EVs stock in the new policies scenario would cut demand for oil products by 127 million tonnes of oil equivalent (Mtoe) yearly; this is about 2.5 million barrels per day in 2030 [14]. The vehicle-to-grid mechanism is undisputedly one of the best steps in integrating EVs into the electric grid [15]. As EVs continue to gain considerable share in the market place, policymakers, utilities, and grid operators are beginning to pay more attention and critically address the management of vehicle charging to smoothen the integration of EV loads on the grid. Different methods have been recommended to shift vehicle charging to low demand periods to ease impacts on the grid, particularly on low-voltage distribution infrastructure [16].

Researchers have proposed that the vehicle-to-grid (V2G) mechanism (with EVs acting as batteries that have bi-directional flows of energy) is the future of EV's integration with the electric grid [17]. Nevertheless, there is a need to balance the requirements, costs, and benefits of V2G among the three primary stakeholders (grid operators, vehicle owners, and vehicle manufacturers). Vehicles must be equipped with the capability to allow two-way flows of electricity between the vehicle and the grid [18]. It is paramount that grid operators are benefactors of the available vehicles in order to reduce the extra cost of controlling and monitoring the V2G interactions, administering the system, and paying owners [19]. Other important aspects of V2G are security and privacy. Regular availability of communications and management for a large network of charge points, vehicles, and user information will require a series of security and privacy concerns. While it is important to prioritize security and privacy in the V2G mechanism, providing routine and periodical maintenance, deliberately ensuring data integrity, and proper monitoring of cybersecurity has been reported to have significant costs [20]. V2G requirements such as minimum plugged-in time and the possibility for occasional loss of vehicles for owners would have to be measured against the needs of utilities and aggregators [21].

Despite the extensive research on RES and its integration for electrification, it is noteworthy that many developing (especially African) countries are still faced with diverse critical electricity crisis. In Zambia, 22% of the total private households are reported to be connected to the national grid [22]. The daily blackouts witnessed in different parts of the country are a result of electricity supply shortages and existing inequalities within the society [22]. Just like other developing countries with a similar electricity crisis, the electric power sector in Pakistan is currently facing the formidable challenges of a suboptimal infrastructure, an insufficient installed capacity, revenue shortage, and circular

debt [23]. It is important to note that the current electricity infrastructure in this country is only accessible to 67% of Pakistanis [23]. Also, over 6 million people were reported to be without electricity access in Nepal [24]. Doli and Pachauri [25] estimated the rural populations without electricity access in developing countries and reported that there is slow progress in the expansion of energy access to households in Sub-Saharan Africa. From their study, they reported that over 398 million rural dwellers are without access to electricity in Sub-Saharan Africa while about 228 million rural dwellers in East Asia and the Pacific are faced with the same challenge [25]. In total, over 1381 million people in developing countries across the globe are reported to be without electricity access [26].

In Nigeria (the seventh most populated nation with 2.64% of the total world population), over 55% of the population lives in abject poverty as the limited energy (electricity) access rate in the country is one of the factors affecting the economy. Also, about 40% of the country's population does not have access to electricity while those with access to electricity supply experience an erratic service from the distribution companies [1]. This shows that Nigeria's growing population is underserved and the current energy demand will rise in the future. Table 1 highlights the difference in per capita electricity consumption of Nigeria with other countries (including some economic indicators based on World Bank Development Indicators). Ugwoke et al. [27] reviewed Nigeria's energy access comprehensively and revealed that there is no standardized framework to synergize the existing methodologies or strategies for improving energy access.

Studies in literature have both accessed and proffered different theoretical solutions to the electricity access situation in developing countries. Al-Sumaiti et al. [28] proposed a probabilistic weather-driven house-based approach for electricity access enablement in developing countries. The proposed method involves the optimal scheduling strategy for electricity supply however, the authors failed to show a clear path to achieving 100% sustainable electrification in developing countries. Fu and Jian [29] reported that the trajectory development state of a developing country is an option to consider when solving the electricity access challenges. Although Bos et al. [30] highlighted the benefits/challenges of expanding grid electricity access in developing countries, their study did not provide a feasible pathway with which the grid electricity production can be expanded. Similarly, Dada [31] presented a study to help understand the challenges and benefits of smart/microgrids' implementation for electricity generation in Nigeria. The study [31] concluded that enabling a smart/microgrid system in the country will only lead to reliability, efficiency, and quality improvement in the electricity grid. Some researchers have also studied the impact of electricity on the economy of developing countries. For instance, the trivariate modeling of the nexus between economic growth, urbanization, and electricity consumption in Nigeria showed that the adjustment of the energy portfolio to meet the urban electricity demand will curtail the adverse effect of urbanization on the economy [32]. In another study [33], the authors advocated for full-scale democratization of the electricity sector in most developing countries. This will empower small/local communities and stand-alone producers to solve their country's electricity crisis [33].

Other existing literatures proposed the sole use of RES only to solve the electricity crisis in developing countries. An analysis of the potential

for cleaner energy recovery and electricity generation from abattoir wastes in Nigeria reported that an estimate of 1040 MWh – 1665 MWh of electricity can be generated from this source of energy [34]. Ogunmodimu and Okoroigwe [35] studied the prospects and challenges of solar thermal electricity in Nigeria. According to their study [35], although there is an allocation for 1000 MW solar thermal electricity installation in Nigeria's energy policy, factors like; fossil fuel price, cost of technology, and fossil fuel subsidy have hampered the actualization of this policy. A model for setting and evaluating the developed RE policies in developing countries was presented in another study [36]. This model can estimate the impacts of unmet electricity demand as well as the cost of electricity provision. Poudyal et al. [24] reported the potency of mitigating the current energy crisis in Nepal with RES. Although their study suggests that RES is crucial to mitigating the present energy crisis, it lacks a clear path/method with which the country's power sector can transition to the use of RES. A proposed pathway to a fully sustainable electricity supply for another developing country in the mid-term future recommended the use of solar photovoltaic and batteries for 100% renewable electricity supply [37]. However, this will drastically reduce the energy security of the power sector as the daily availability of solar radiation can either make or break the country's economy/power system in a short period. The study about the transition towards sustainability using renewable energies to solve the Venezuelan energy crisis [38] only focused on policy implementation without a comprehensive techno-economic feasibility study.

Although efficiency improvement and development of new RE-based technologies is the state-of-art for energy research application in developed countries, the efficient integration and utilization/adoption of the existing technologies is still a major challenge in most developing countries. Also, access to electricity is one of the main challenges facing many developing (especially African) countries. As reviewed in the preceding paragraphs, existing literatures have studied the electricity crisis from various perspectives, nevertheless, this problem has lingered in many developing countries till date. The recent climate change effects experienced globally brought about different rules and caps on greenhouse gas emissions by individual countries. These rules and laws have further reduced the growth of the electricity sector for most developing countries as the environmental sustainability of any proposed electricity expansion project are now critically scrutinized. Most of the recent studies in literature have neglected the use of fossil fuel energy sources for electricity generation due to the carbon emission attached to this method. However, they fail to justify the smooth transition of the power sector (dominated by natural gas power plants) to a 100% RE-based sector. Also, the potency of RES for electricity generation is mostly considered from a policy and techno-economic perspective without an in-depth study on the daily/hourly production of the RE-based power system. Furthermore, the simultaneous analysis of the integration of different energy sources to meet the hourly electricity demand in developing countries is limited in the existing literature. One of the main challenges with RES (excluding hydro and biomass) is their intermittency in nature and the inability to provide power supply for 24 h without a storage system.

Therefore, there is a need to develop an economic and effective method to strategically integrated these resources for stable electricity production. In this paper, a comprehensive method is proposed for the

Table 1
World Bank Development Electricity and Economic Indicators [27].

Country	Population with Access to Electricity	GDP Per Capita	GDP	Electricity Consumption
USA	100%	\$ 57807.7	\$ 18.62×10^{12}	12.83 MWh/Capit
Italy	100%	\$ 31,279.1	\$ 1.86×10^{12}	5.1 MWh/Capita
China	100%	\$ 7993.1	\$ 11.22×10^{12}	4.05 MWh/Capit
India	82%	\$ 1706.5	\$ 2.26×10^{12}	0.86 MWh/Capit
South Africa	86%	\$ 5274.3	\$ 2.95×10^{11}	4.15 MWh/Capit
Nigeria	61%	\$ 2175.7	\$ 4.05×10^{11}	0.14 MWh/Capit

efficient utilization of RES for electrification in developing countries. The materials and methods presented in this study are developed to enhance the large integration of renewables into the power systems for electricity while considering hydrogen production and electric vehicle integration for clean electricity maximization. The proposed method is aimed at bridging the aforementioned gaps in existing literatures. Although this method is adopted/proposed for developing countries, it can also be used by developed nations as a template for the smooth transition from non-RE electrification to 100% RE-electrification. The novelty and contributions of this study are highlighted as follows.

- The development of a model that strategically integrates RES for stable electricity production. The sustainable integration of these resources with (existing/new) fossil fuel systems for large-scale electricity generation is proposed to reduce carbon emissions.
- This study presents a pathway with which developing countries can achieve sustainable electrification by considering financial feasibility, environmental sustainability in terms of CO₂ reduction, and technical viability.
- In comparison to existing literature that considered the use of RES only, the integration of both RES and non-RES is analyzed in this study. The integration of plug-in EVs and hydrogen production into the electricity generation mix is further analyzed to actualize RE penetration and to maximize the RE electricity produced.
- Development of a novel method/plan with which RES and non-RES are used to solve electrification crisis/challenges. Analysis of RES viability considering its hourly electricity generation potential.
- Integration of EVs and hydrogen production for 100% stability of RE-based electricity generation network and maximization of the produced electricity. The determination of feasible and infeasible pathways for electricity generation with RES and non-RES combinations.

It is noteworthy that this is the first literature that proposes a model that integrates the use of RES and non-RES for the large-scale electricity planning of a country while prioritizing economic feasibility and environmental sustainability. Using Nigeria as a case study, an electricity demand/supply analysis based on an hourly time-step is done in a bid to solve to electricity generation crisis in this country. The estimated electricity demand of the country by 2030 is considered in this study as this year coincides with the country's long-term target to solve its electrification problem [39]. Three renewable energy sources (hydro, wind, and solar) and one fossil fuel (natural gas) type are considered within the scope of this study. The fossil fuel source is chosen based on its availability and its current applicability in the electricity sector in the case study. It is noteworthy that over 60% of the electricity generation in Nigeria currently is from natural gas power plants (NG). Furthermore, the integration of five RE-based technologies namely; offshore wind power plant (WindOFF), onshore wind power plant (WindON), solar photovoltaic systems (SolarPV), concentrated solar power plant (CSP), and hydropower plant (Hydro) are considered within the scope of this study. These have been considered based on their maturity in application globally and the availability of raw materials/resources. The pumped hydro storage system (ST) is the only storage system used in the simulation as it is the most (environmentally) sustainable storage system. Also, there are a lot of sites for ST installation in this country. In most literature, the use of solar and wind energy is considered without a strategy to maximize the electricity produced from these systems. The integration of plug-in EVs and hydrogen production into the electricity generation mix is further analyzed to actualize RE penetration into Nigeria's electricity sector and also to maximize the RE electricity produced. It is noteworthy that biomass energy resource is not considered in this study as the data about the potential of this energy source is inconsistent and unavailable in some cases. Also, geothermal, tidal, and some other RE sources have been exempted in this study due to their applicability, technicality, and availability of raw materials. This study

can be used or replicated by (developing or developed) countries with similar electricity generation (as well as environmental sustainability) challenges. The proposed method for this study is justified in [Section 2](#) while the results of the analysis and its discussions are detailed in [Section 3](#). [Section 4](#) highlights the main concluding points from this study.

2. Materials and methods

In this study, the use of different RES and natural gas, as well as EVs integration, is analyzed with the aim to solve an imminent electrification challenge in Nigeria. The production of hydrogen from the generated excess electricity by the RE-based technologies is implemented and analyzed to enhance RE penetration into the power sector. The method proposed in this study is implemented in EnergyPLAN model and MATLAB programming environment. In this section, the materials and method adopted and the data used for the simulations in this study are justified. A brief introduction to the EnergyPLAN model is first presented while the model developed is validated considering the current situation of Nigeria's power sector. The modeling of the renewable energy system is also presented while the process of integrating hydrogen production and EVs is explained. The analysis presented in this study has been done based on the following assumptions:

- The dynamic response of the power system or power network to different faults is neglected.
- The transmission and distribution network is sufficient for the power generated and losses from these networks are neglected.
- Financial constraints are not considered when modeling the capacities of the power plants for different scenarios.
- Economic feasibility and financial analysis are done based on the new installation of all the power plants.
- All the analysis has been made assuming a constant load profile.

In this study, we first analyze Nigeria's current electricity generation/supply scenario. This will further justify the need for 100% electrification by 2030 and also validate the adopted model in this study.

2.1. EnergyPLAN model

EnergyPLAN is a commercial software/program that was developed at Aalborg University and it incorporates the economic, technical, and environmental aspects of energy systems in enabling the optimization of different mix/topology of energy technologies [40]. The program has a platform for integrating the electricity, heat, and transport sectors of a particular case study. It is effective in describing future energy systems, as it allows for comparative analysis of different energy systems and simulates regional and national energy systems on an hourly basis. The EnergyPLAN model utilizes several criteria in assessing and calculating the best energy system (technology) mix for a specified location and period. Some of these criteria are [41]:

- *Primary energy supply*; which is the sum of the fuel usage and fuel equivalents for non-fuel energy sources.
- *Carbon dioxide emissions*.
- *Excess power generations*; this gives information about the ability of an energy system to be implemented in a system with fluctuating power sources. This is also referred to in terms of exportable excess electricity.
- *Socio-economic costs and business economic costs*; which include the total annual costs, fuel costs, and operation and maintenance costs.
- *Renewable energy share and fossil fuel share*.
- *Production mix* across primary energy sources

The goal of the EnergyPLAN model is to highlight the optimum technology mix for configuring the energy system. This is done by assessing pre-defined energy system configurations. Furthermore, it

helps to ensure the strategic design of large energy systems through techno-economic analysis of diverse configurations. This can be applied on different levels including towns and municipalities [42], regional level, national level [41], as well as a continental level [43]. EnergyPLAN has been utilized by researchers in addressing local areas such as the Southwest Region in Ireland, municipalities in Mljet in Croatia, and Aalborg in Denmark. It has also been used for the energy planning of towns like Corinaldo in Italy and Loviisa in Finland, as well as other large-scale systems in the US, China, Oceania, and Europe. The application of EnergyPLAN in articles spans through different scopes. They include integration of RES in energy systems [44], transport [45], grid stability [46], biomass usage [47], district heating [48], desalination [49], and transmission issues [50]. The EnergyPLAN layout is illustrated in Fig. 1 and this program is used to implement the sustainable strategy proposed in this present study. As seen in Fig. 1, the inputs for this research include; electricity/EV demand profile, RE profile, RE technologies, capacities and efficiencies, storage system, and simulation regulations.

2.2. Electricity demand estimation

The dynamics of electricity demand in Nigeria are attributed to population size, income, and industrialization [51]. In the recent past (between 2010 and 2014), the per capita electricity consumption in Nigeria was at an average of 146 kWh [52]. It was also reported that Nigeria registered a low energy consumption of about 129.6 Mtoe, which was caused by fuel scarcity during high-demand periods. The electricity demands have continued to grow and this is attributed to increasing development and population growth [27]. Based on a study [53], the total yearly electricity demand in Nigeria is expected to increase from 35.9 billion kWh in 2010 to a projected 283.6 billion kWh by 2040. The study explained that this increase will be proportional to urbanization and income growth rate. The estimated electricity load

according to a previous study [54] was 16,500 MW. The inaccuracy in the electricity demand projections in these previous studies [53–55] can be attributed to the lack of credible data in the country's power sector. Also, the consideration of the different variations in load demands during the estimated periods is another factor that affects the variation in projected demand. Two studies [53,56] in the existing literature that worked on the projected electricity demand in Nigeria reported 19.6 GW to be the predicted national grid load by 2030. However, in order to accommodate excesses and taking into consideration the uncertainties attached to the two studies, this present study assumes a total annual electricity demand of 200 TWh/yr for Nigeria by 2030. The simulation strategy used in this study incorporates the hourly load profile based on the 200 TWh/yr estimated electricity demand.

2.3. Research modeling

In this study, the feasible pathways to achieving daily stable (100%) electrification in a developing country are simulated and analyzed using 2030 as the targeted year. It is worth noting that the Nigerian government has set an electrification target of 30 GW on-grid capacity, for which 30% will come from RES [57]: therefore, this study gains significance as it would be instrumental in assessing the practicality of these electrification targets (based on the generation side). Both RES and non-RES are proposed for electricity production. The RE and non-RE technologies considered are offshore wind power plant (WindOFF), onshore wind power plant (WindON), solar photovoltaic systems (SolarPV), concentrated solar power plant (CSP), Pumped hydro storage system (ST), hydropower plant (Hydro), and natural gas power plant (NG).

This study also focuses on environmental sustainability in terms of carbon emission reduction, therefore, concerns about the environmental effects of the technologies are assessed. The EnergyPLAN simulation program takes into consideration the technical details of each

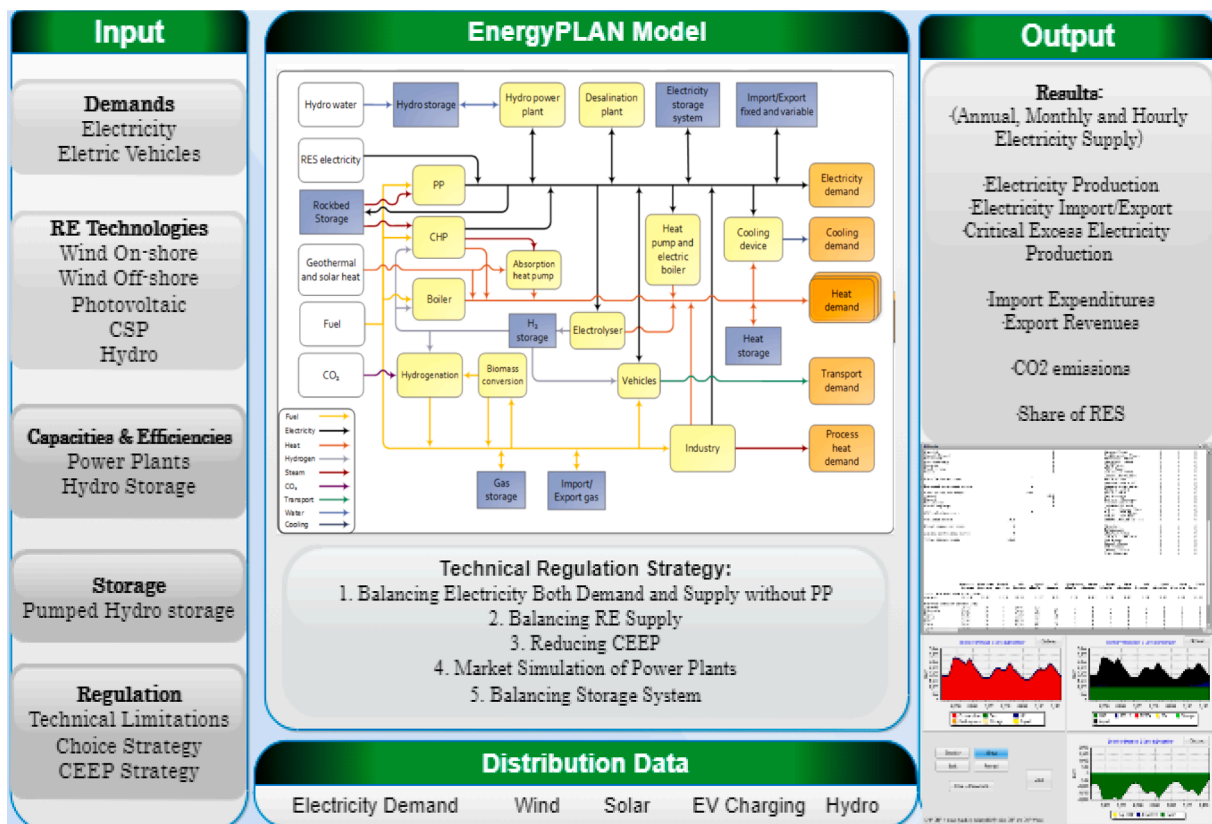


Fig. 1. Summary of EnergyPLAN simulation program.

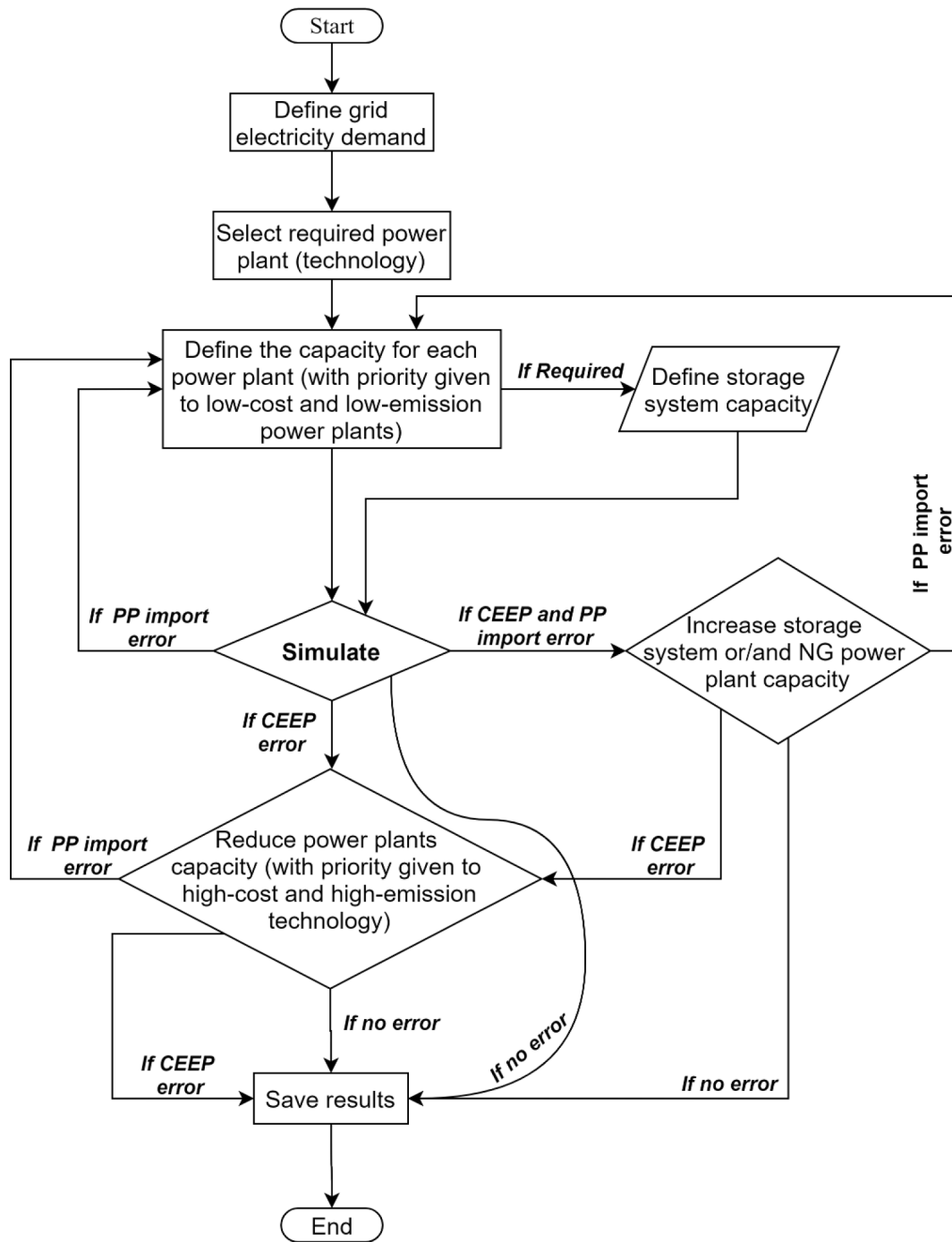


Fig. 2. Flow chart of the simulation procedure for the proposed method.

technology utilized in this study. Each of the energy technology (with the exception of ST) was first simulated to meet the predicted electricity demands. Then, the combination of these technologies is simulated and analyzed. The total number of scenarios considered in this study is 99 and the most efficient solution (for different combinatory levels) is ascertained based on some specific criteria. Fig. 2 shows the simulation flow chart as well as the optimization procedures for the proposed method. Considering environmental sustainability, priority was given to the RES technologies in meeting (a larger share of) the 200 TWh/yr demand.

As seen in Fig. 2, there are four (simulation possible) outcomes based on each defined supply for each scenario. These are explained as follows:

- **No (CEEP and PPI) error:** in this case, the energy supply balances the electricity demand. For optimal supply, a reduction in supply is applied (with the NG being the priority due to the focus on more RES integration). Also, the reduction of the most expensive RES is done for economic reasons.
- **PPI error:** This warning shows that there is a need for more energy supply of power due to the insufficient power production to meet hourly demands throughout the year. While increasing the energy supply, the priority is RE technologies until they reach their maximum potential. Also, NG is supplied if the warning persists.
- **CEEP and PPI error:** This signifies an insufficiency in power production (for some hours) as compared with electricity demands.

Despite the excess CEEP by the RE sources, more NG/ST capacity is defined and the RE sources are reduced.

- **CEEP error:** In this scenario, there is excess power production or excess electricity in the power network. This situation usually occurs due to excess electricity produced by the RE sources, as the NG supply stops when demand is met. In this case, the reduction of RE production and utilization of hydro-storage are methods applied in this study.

2.4. Input data and technological definition

In order to have a realistic simulation, the technological definitions for each of the RE-technology must be done carefully. In this subsection, the input parameters used in the simulation and analysis of all the systems are justified. Also, each technological definition is explained extensively.

2.4.1. Solar energy technologies definition/modeling

Nigeria being a tropical country lies between 4° and 13°, and it enjoys over six hours of sunshine daily. The coastal region in the northern part of the country can have as much as 9–13 h of sunshine at different periods of the year. Also, the average solar radiation at the coastal area and northern boundaries are between 3.5 kWh/m²/day and 9.0 kWh/m²/day respectively [58]. These values are within the recommended threshold range of 1900 kWh/m²/yr–2100 kWh/m²/yr by CSP technology manufacturers. The electric potential of solar PV technology in Nigeria is estimated to be above 1000 kWh/kWp [59]. The location used in this study for retrieving the solar insolation data is 9.142 longitude and 10.182 latitudes. The solar insolation data for this location is used for the simulation. The EnergyPLAN considers the technical details of the SolarPV and CSP. Storage devices were not considered in the PV and CSP modeling process. The only storage system used in this study is the hydro-storage due to its viability/sustainability in Nigeria. Due to the last expanse of land in Nigeria and the enormous potential of solar PV or CSP, the possible maximum installed capacity for these technologies is limited to 100,000 MW. This assumption is based on the feasibility study carried out in the literature [60].

2.4.2. Wind energy technologies definition/modeling

Based on existing works of literature [61,62], the prospect of wind power is relatively high in Nigeria. According to the Nigerian Meteorological Service, the Northern parts of Nigeria (especially the area north of latitude 10°) are windier than the Southern parts. It is estimated that the annual mean wind speed in the Northern parts is above 2.5 m/s, while it is lower in the southern part [63]. The northern region of Nigeria has capacity factors between 11% and 45% [63]. Across the 6 geographical zones in Nigeria, the most viable regions for grid-integrations are the North-Central, North-West, and North Eastern regions while the least attractive for is the south-west and south-East regions. In this study, simulations of onshore/offshore wind turbines were made based on real-life wind turbine applications. Furthermore, due to the large expanse of land space and sea area in Nigeria, over 100,000 MW WindOFF and WindON installations are possible. The EnergyPLAN was preprogrammed to simulate a 1 MW wind turbine system for onshore/offshore applications, with consideration of losses and other technical factors affecting its production. A typical wind profile over a period of one year is used for the simulation in this study.

2.4.3. Hydro-power and pumped hydro-storage

Over 20% of Nigeria's electricity is currently generated via hydro-power and has been confirmed by other studies [64]. The major hydropower stations in Nigeria are Kainji (760 MW), Jebba (578.40), and Shiroro (600 MW). Based on literature, the total (including large, small, and micro) hydropower potential is between 13,000 MW and 14,750 MW [65]. For accuracy purposes, the maximum hydropower potential is taken as 13,000 MW for this study. Nigeria has a hydropower installed

capacity of 1975 MW. In this present study, the simulation was done using the hydro dam profile of a typical river in Nigeria. The phenomenon of storing energy using ST involves pumping water uphill to the upper reservoir. ST is an efficient storage technique for storing excess electricity produced by renewable energy technologies (wind turbines, CSP, SolarPV). The advantages of utilizing ST are low cost, high round-trip efficiency, less land (off-river ST) [66]. Pumped hydro storage systems constitute about 160 GW of global stored power and over 99% of stored energy.

2.5. Economic feasibility and emission analysis

The economic feasibility report is an important factor used in assessing the viability of an electricity generation and other energy systems' project. This study employs the EnergyPLAN cost database for 2030, in estimating the economic parameters, i.e. initial investment costs and operation and maintenance (O & M) costs. The economic analysis also considers the fixed and variable O & M costs. The fixed costs in the project comprise the recurring yearly costs, while the variable O & M costs are dependent on the amount of energy used. Fuel costs constitute a significant aspect of the variable O & M costs in the NG-based power plants. A 3% interest rate is used in the economic analysis in this study. The expected expense of the power plants in 2030 based on the current price and the carbon emissions of each scenario were calculated. This ensured that an accurate forecast of possible emissions from the utilization of NG is made. The emission factor of the NG used in this study is 56.9 kg/GJ of electricity produced based on literature [40,67]. Table 2 presents a summary of the financial parameters for the technologies considered within the scope of this study. The economic analysis in the study is done by considering the new installation of all the power plants (i.e., the cost reduction based on existing power plants is neglected).

2.6. Mathematical modeling

In the section, the equations used in modeling the RE technologies and making economic analysis are presented. Typically, the design of a SolarPV consists of calculating the appropriate number of PV modules that are required to convert solar energy into direct current (DC) power. The output power of each PV module at hour t ($p_{PV(t)}$) can be modeled with Eq. (1):

$$p_{PV(t)} = I(t) \times A \times \eta_{PV} \quad (1)$$

where $I(t)$ denote the hourly solar insolation (kW/m²), η_{PV} is the

Table 2
Power plants economic parameters [68–71].

Type of Power Plant	Lifetime (Years)	Fixed O&M (% of Investment)	Investment Cost (M \$/unit)	Fuel Cost (\$/GJ)	Variable O & M Cost (\$/MWh-e)
NG (MWe)	27	3.16	0.98	10.2	
TurbineST (MWe)	50	1.5	0.6		1.19
PumpST (MWe)	50	1.5	0.6		1.19
ST Capacity (GWh)	50	1.5	7.5		1.19
CSP (MWe)	8.2	25	5.98		
SolarPV (MWe)	40	1	0.82		
Hydro (MWe)	50	2	3.3		1.19
WindOFF (MWe)	25	2.94	2.43		
WindON (MWe)	25	2.59	1		

efficiency of the PV module and A denotes the PV module area (m^2). In this study, losses in PV modules are also considered as this has been pre-program in the EnergyPLAN model.

The WindOFF and WindON are modeled considering all the technicalities of a working wind power plant. As the wind speed rises above the cut-in velocity, the power delivered by the generator is proportional to the cube of wind speed. Once the speed surpasses the rated velocity (V_r), the power generated by the generator is constant and equal to the rated power (P_r). If the cut-out value (V_{co}) is exceeded, the machine stops operating to preserve the generator, resulting in zero output power. Mathematically, the power generated by each wind turbine (p_{WT}) at hour t is modeled with Eq. (2):

$$p_{WT}(t) = \begin{cases} 0 & v(t) \leq V_{ci} \text{ or } v(t) \geq V_{co} \\ P_r \cdot \frac{v^3(t) - V_{ci}^3}{V_r^3 - V_{ci}^3} & V_{ci} < v(t) < V_r \\ P_r & V_r < v(t) < V_{co} \end{cases} \quad (2)$$

For the River hydropower plant (Hydro) the power production from incoming water is calculated with Eq. (3) [72].

$$P_{total} = Q \times \rho \times g \times H \times \eta \quad (3)$$

where Q denotes the flow rate in the pipe (m^3/s) and H represents the waterfall height or head (m). g is the acceleration due to gravity (m/s^2) and ρ represents the water density (kg/m^3). η and P_{total} represent global efficiency ratio and total power producible.

The economic analysis in this research will focus on the investment and running cost as stated in Section 2.5. The total investment (I_{tech}) of each production unit is calculated as:

$$I_{tech} = C_{tech} \times P_{unit-techn} \quad (4)$$

where $P_{unit-techn}$ is the per-unit price and C_{tech} is the total capacity of the technology. The total annual cost ($A_{TOC-techn}$) for each technology is calculated as:

$$A_{TOC-techn} = A_{investment-techn} + A_{FOC-techn} + A_{VC-techn} \quad (5)$$

where $A_{investment-techn}$ is the annual cost for each of the technology or technological combinations (scenarios) and this is modeled with Eq. (6). $A_{FOC-techn}$ represents the annual fixed operational costs and is modeled with Eq. (7). $A_{VC-techn}$ is the variable cost.

$$A_{investment-techn} = I_{tech} \times \frac{i}{[1 - (1 + i)^{-n}]} \quad (6)$$

where n is the total lifespan of each technology on years, i is the interest (and this is 3% for this study).

$$A_{FOC-techn} = P_{FOC-techn} \times I_{tech} \quad (7)$$

where $P_{FOC-techn}$ is the annual fixed O & M costs and it is modeled in percentage of the investment cost.

2.7. Hydrogen production

Hydrogen production from electricity can be achieved through different processes, however, its production from renewable electricity via power-to-gas process (PTG) bears the greater advantage. It complements the intermittent power supply from solar and wind energy installations [73]. It is noteworthy that the economics and financial commitment for the electrolytic production of hydrogen have so far been regarded as too expensive [74]. Considering a hybrid of RES with a PTG facility that generates hydrogen from electricity and water, at each point in time, the electricity generated can be sold easily. The mathematical equation for the PTG considering a polymer electrolyte membrane (PEM) is summarized in Table 3. In accordance with Glenk and Reichelstein's study [73] that was recently published in Nature Energy Journal, the PTG conversion rate of 0.019 kg/kWh is adopted in this study.

Table 3

The equations used in modeling hydrogen production [75]

Definitions	Equations
The chemical equation for hydrogen production	$\text{H}_2\text{O} + 2\Delta\text{H} \rightarrow 2\text{H}_2 + \frac{1}{2}\text{O}_2$
Anode reaction	$\text{H}_2\text{O} \rightarrow 2\text{H}^+ + 2\text{e}^-$
Cathode reaction	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
Mass balance	$\dot{m}_{\text{H}_2\text{O}} = \dot{m}_{\text{H}_2} + \dot{m}_{\text{O}_2}$
Energy Requirement	$\Delta G = \Delta H + T\Delta S$
The molar flow rate of hydrogen produced	$\dot{N}_{\text{H}_2} = \frac{J}{2F}$
Electrical energy requirement for electrolyzer	$E_{\text{elec}} = Q_{\text{elec}} = JV$
Electrical overpotential	$V = V_0 + V_{\text{ohm}} + V_{\text{act,a}} + V_{\text{act,c}}$
Ohmic of electrolyte	$V_{\text{ohm}} = JR_{\text{PEM}}$
Cathode and anode activation overpotential	$V_{\text{act,i}} = \frac{RT}{F} \sinh^{-1} \left(\frac{J}{2J_{0,i}} \right) i = a, c$
Cathode and anode current density	$J_{0,i} = J_i^{\text{ref}} \exp \left(-\frac{E_{\text{act,i}}}{RT} \right) i = a, c$
Reversible potential	$V_0 = 1.229 - 8.5 \times 10^{-4} (T_{\text{PEM}} - 298)$
The overall ohmic resistance of the electrolyzer	$R_{\text{PEM}} = \int_0^D \frac{dx}{\sigma_{\text{PEM}}[\lambda(x)]}$
Ionic conductivity of electrolyzer	$\sigma_{\text{PEM}}[\lambda(x)] = [0.5139\lambda(x) - 0.326] \exp \left[1268 \left(\frac{1}{303} - \frac{1}{T} \right) \right]$
Electrolyzer water content at a distance x from the anode and cathode	$\lambda(x) = \frac{x\lambda_a - x\lambda_c + \lambda_c}{D}$

2.8. Electric vehicles integration

Since EVs are already in use in many countries and will be in wide application globally soon, the integration of this (EVs) system into a future planned power/energy network is paramount. Although, there are a lot of EV emerging technologies such as vehicle-to-vehicle and vehicle-to-grid operations [76], plug-in battery electric vehicle (BEV) is the only system considered within the scope of this study. The people's republic of China is the world's largest EV user with over 1.77 million EVs integrated with their transport system and power network [77]. This study is modeled following the operations of plug-in BEVs in China. Based on research [78,79], most EVs are charge between 6 and 12 h per day (from late evening hours to early morning hours). In this study, the plug-in EVs are designed to charge for 12 h daily per day using a China-based EnergyPLAN profile. This does not necessarily mean each EV will be charged throughout this period, however, it implies that the power systems will be open for EV charging during this 12-hour window.

2.9. Model validation

In order to validate and the accuracy of the simulation tools, model, strategy, and method proposed in this study, the analysis of the current power systems in Nigeria (in terms of generation) is done. Also, This will focus on Nigeria's electricity demand and supply only. According to [57], the available electricity capacity is 6056 MW, from an installed capacity of 10,396 MW. The present electricity demand in Nigeria is approximately 144.5 TWh/yr [80]. Furthermore, the available hydro-power capacity and electricity produced are 1060 MW and 5.0 TWh/yr respectively. These values are used to analyze the current power system and Fig. 3 shows the current yearly electricity load profile for Nigeria, which is used in the analysis of the present scenario. From an available thermal power plant capacity of 4996 MW, the electricity generated is 43.88 TWh/yr. The electricity produced by two types of plants (43.88 TWh/yr and 5.0 TWh/yr) is 33.84% of the required demand. Fig. 3 shows the annual electricity produced against electricity demand. The insufficient access to 24 h electricity by Nigerians is explained by the 25% reduction in the (59%) quoted access to electricity [81]. Furthermore, this suggests that there are barely 13–14 h of electricity usage to the (59%) population quoted [82] to have access to

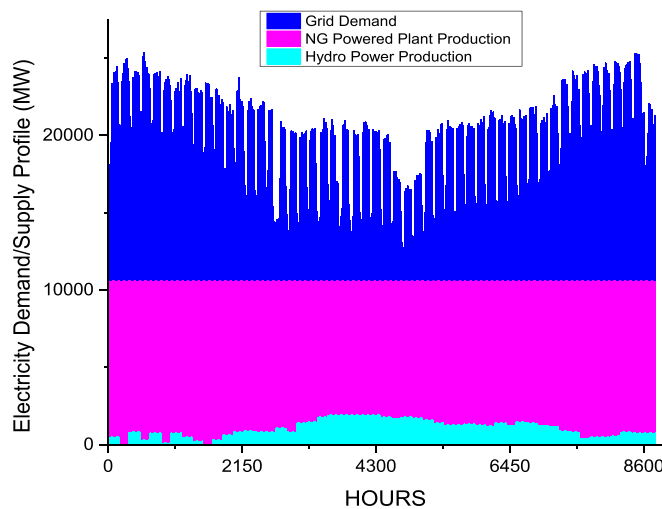


Fig. 3. Model validation result.

electricity. This matches the results presented by previous studies in literature. This validates the model, simulation program, materials, and method presented in this study and therefore can be used for future predictions.

3. Results and discussions

In this study, a strategic plan to solve the imminent energy poverty (inadequate electricity generation) problem for the most populated African country is presented. The integration of different renewable energy resources to solve this problem as well as the use of natural gas power plants is analyzed. The role of hydrogen production and BEV integration in the stability of electricity production and reduction of electricity wastage is also studied. In this section, the results from the combination of different resources/technologies are presented. This is done by highlighting different feasible scenarios for the different case scenarios in this study (Section 3.1). For simplicity reasons, the combination of these renewable and non-renewable sources, as well as the storage system, is first analyzed. After which the integration of BEVs and hydrogen production is considered for the optimal scenarios in each combination group and the results are discussed in Section 3.2. It is however noteworthy that the analysis presented in this study has only considered the generation side of the electricity network while transmission and distribution details are not explicated. In comparison to existing studies, the electricity demand and supply planning presented in this study is based on an hourly time-step. This is done with the aim to use the available resources and technology to meet all the electricity demand every hour.

3.1. Overview of electricity production from different technological combinations

Considering the simulation strategy presented in the preceding section (Section 3), the results of each feasible case for the ninety-nine combinatory scenarios in this study are highlighted in Table 4. This table highlights the technological combinations, the required installed capacity for different technology, the total annual carbon emission from electricity generation, the required total investment, the annual cost (which includes the operation and maintenance cost as well as yearly investment cost), the share of renewable energy systems (RES) in the primary energy supply (PES), the share of RES in electricity production, critical excess electricity production (CEEP), and yearly electricity production by RES. The most feasible solution will be highlighted for the use of one, two, three, four, five, six technologies as well as the combinations of all technologies considered within the scope of this study.

The sole use of all the renewable energy technologies considered within the scope of this study is incapable of meeting the hourly electricity requirement of our case study. This is due to the obvious intermittent nature exhibited by wind and solar energy resources. Also, the combination of two or more RE-based technology without a storage system is not feasible in meeting the hourly electricity demand. The integration of a storage system or the use of an NG power plant will be required to augment the production by the RE-based power plants.

The use of NG energy resources is the only feasible solution if the area of study is to adopt the generation of electricity from a single technology. This will require a 36,000 MW installed capacity and it will emit 102.06 Mt of CO₂ yearly. The feasibility of this electricity production plan is low as this will reduce the energy security of the country. Although Nigeria is a high prosumer of natural gas, the carbon emission rate attached to the utilization of this technology for electricity production poses a more complex problem for the country in the future. Based on the total investment cost, the production of electricity from NG and WindON is the most feasible dual technology solution (Table 4). However, the use of NG and Hydro is more feasible considering the total annual cost (\$ 19.2B). Currently, Nigeria generates electricity from NG and Hydro resources and if the country will continue in this direction, the capacity of hydropower will have to increase to 13,000 MW by 2030. Also, the installation of a total of 28,800 MW of NG power plant is required to meet the estimated electricity demand. Although the use of this technology combination will not result in critical excess electricity production (CEEP), the yearly carbon emission is significantly high (77.27 Mt). The integration of the hydro storage system (ST) with SolarPV or WindON results in the most environmentally friendly scenario considering its futuristic sustainability (Table 4). These two scenarios will be further explored in Section 3.2. It is worth mentioning that the adoption of the dual technologies WindOFF/ST or CSP/ST is not a feasible solution economically as their initial investment cost is too high (\$ 601B for WindOFF/ST and \$ 680B). Technically, the high ST capacity required is also a concern as well as the sole dependence on an intermittent RES.

Considering the use of three different technologies, the use of RE-based technology requires either a storage system (ST) or NG to achieve electricity production with 100% stability. Hydro technology is one of the few renewable energy technologies that can be used to meet based load demand without the need for ST, however, the available hydropower resources in Nigeria are insufficient in comparison to the estimated grid electricity demand by 2030. Considering costs, the integration NG, WindOFF, and SolarPV offer the cheapest solution with an initial investment cost of \$ 57.7B and an annual investment cost of \$ 22B. This scenario will emit 76.25 Mt of CO₂ yearly which will significantly increase the overall CO₂ emission of the country. Although there will be critical excess electricity production, the share of RES in the total PES (11.9%) is significantly low so also, the annual electricity produced by RES (50.58 TWh/yr). The most environmentally friendly solution using tri-technology for electricity production is the combination of WindON, ST, and SolarPV. This offers the country a means to generate a CO₂ emission-free electricity but there will be some excess electricity production from the RES power plants. The total investment cost for this scenario is \$ 178B and the annual total cost is \$ 11.44B (Table 4). With the rise in awareness for energy diversity, energy security, and environmental sustainability, the use of NG, Hydro, and SolarPV will satisfy these criteria. Also, the adoption of this scenario will reduce the total investment (\$ 79.73B) and total annual cost (\$ 19.61B) as the use of NG and Hydro technologies are already in existence in Nigeria. The share of RES in PES is 20% and the share of RES in total electricity production 38.4%.

The integration of four of the different technologies considered within the scope of this study provides the highest number of feasible, technically viable, and cost-effective options. It either involves the combination of NG powerplants with RES powerplant or the use of ST to stability RES production. As seen in Table 4, the total investment and

Table 4

Simulation results summary for different technological combinations.

Technology Combination	Generation Capacities	Annual Carbon Emission (Mt)	Total Investment (B\$)	Total Annual Cost (B\$)	RES Share in PES (%)	RES in Electricity Produced (%)	CEEP	Annual Electricity Production by RES (TWh/yr)
Generation from single and two technologies								
NG	36,000 MW	102.06	35.3	26.2	0	0	NO	0
NG, WindOFF	32,000 MW NG; 14,000 MW WindOFF	88.23	65.4	24.3	8.8	19.4	NO	38.85
NG, WindON	34,500 MW NG; 7,500 MW WindON	93.45	41.1	24.7	3.6	8.4	NO	16.88
NG, Hydro	28,800 MW NG; 13,000 MW Hydro	77.27	71.1	19.2	35.4	46.4	NO	92.82
NG, SolarPV	33,000 MW NG; 13,000 MW SolarPV	80.47	43	23.2	6.2	14.2	NO	28.92
NG, CSP	33,000 MW NG; 11,000 MW CSP	80.66	98.1	32.1	5.7	13.1	NO	26.27
SolarPV, ST	90,000 MW SolarPV; 2000 GWh ST; 48,000 MWe PumpST; 36,000 MWe TurbineST	0	279.4	15.39	100	74.6	Yes	241.52
WindOFF, ST	75,000 MW WindOFF; 12,000 GWh ST; 18,000 MWe PumpST; 31,000 MWe TurbineST	0	601	22.34	100	89	Yes	208.14
CSP, ST	100,000 MW CSP; 3500 GWh ST; 58,000 MWe PumpST; 36,000 MWe TurbineST	0	680	88.1	100	71	Yes	232.07
WindON, ST	100,000 MW WindON; 22,000 GWh ST; 38,000 MWe PumpST; 35,000 MWe TurbineST	0	309	19.76	100	81.1	Yes	231.19
Generation from three technologies								
NG, WindOFF, WindON	31,000 MW NG; 9,000 MW WindOFF; 11,500 MW WindON	75.75	63.8	22	12.2	25.8	NO	51.56
NG, WindOFF, Hydro	27,000 MW NG; 6,500 MW WindOFF; 13,000 MW Hydro	68.45	85.2	21.71	16.4	32.9	NO	65.87
NG, WindOFF, SolarPV	31,000 MW NG; 6,500 MW WindOFF; 14,000 MW SolarPV	76.25	57.7	22	11.9	25.3	NO	50.58
NG, WindOFF, CSP	31,000 MW NG; 12,000 MW WindOFF; 7,000 MW CSP	76.53	102	28.4	11.8	25.0	NO	50.02
NG, WindON, Hydro	27,500 MW NG; 2,500 MW WindON; 13,000 MW Hydro	74.70	72.4	22	12.8	26.8	NO	53.61
NG, WindON, SolarPV	32,500 MW NG; 7,000 MW WindON; 12,000 MW SolarPV	80.16	48.7	22	9.9	21.5	NO	42.91
NG, WindON, CSP	32,500 MW NG; 11,500 MW WindON; 8,500 MW CSP	78.13	94.2	28.52	10.9	23.4	NO	46.89
WindON, CSP, ST	75,000 MW WindON; 30,000 MW CSP; 4,000 GWh ST; 31,000 MWe PumpST/ TurbineST	0	322	35.02	100	93.2	Yes	245.04
NG, Hydro, SolarPV	28,800 MW NG; 13,000 MW Hydro; 1,000 MW SolarPV	79.57	71.95	23.05	11.4	24.3	NO	50.07
NG, Hydro, CSP	26,500 MW NG; 13,000 MW Hydro; 9,000 MW CSP	62.42	143.6	29.35	20.3	38.8	Yes	77.69
NG, SolarPV, CSP	31,000 MW NG; 50,000 MW SolarPV; 9,000 MW CSP	53.99	125.2	24.56	33.4	66.4	Yes	132.86
Hydro, CSP, ST	71,000 MW CSP; 13,000 MW Hydro; 27,000 GWh ST; 43,000 MWe PumpST; 23,000 MWe TurbineST;	0	710	74.94	100	77.7	Yes	217.41
SolarPV, CSP, ST	25,000 MW CSP; 75,000 MW SolarPV; 23,000 GWh ST; 41,000 MWe PumpST; 30,000 MWe TurbineST;	0	426	35.93	100	88.1	Yes	265.98
WindOFF, SolarPV, ST	30,000 MW WindOFF; 78,000 MW SolarPV; 15,000 GWh ST; 36,000 MWe PumpST; 26,000 MWe TurbineST	0	287	17.93	100	97.4	Yes	254.2
WindON, Hydro, ST	85,000 MW WindON; 13,000 MW Hydro; 19,000 GWh ST; 21,000 MWe PumpST; 28,000 MWe TurbineST	0	300	19	100	93.1	Yes	244.35
WindOFF, WindON, ST	30,000 MW WindON; 59,000 MW WindON; 15,000GWh ST; 16,000MWe PumpST; 26,000MWe TurbineST	0	270	18.77	100	93.0	Yes	230.99
Hydro, SolarPV, ST	79,000 MW SolarPV; 13,000 MW Hydro; 29,000GWh ST; 43,000MWe PumpST; 24,000 MWe TurbineST	0	365	20.03	100	78.6	Yes	220.97
WindOFF, CSP, ST	35,000 MW WindOFF; 50,000 MW CSP; 21,000 GWh ST; 27,000 MWe PumpST; 26,000 MWe TurbineST	0	573	59.39	100	84.5	Yes	216.55
WindOFF, Hydro, ST	57,000 MW WindOFF; 13,000 MW Hydro; 14,000 GWh ST; 27,000 MWe PumpST; 25,000 MWe TurbineST	0	318	21.96	100	89.1	Yes	206.02
WindON, SolarPV, ST	60,000 MW WindON; 75,000 MW SolarPV; 4,000 GWh ST; 29,000 MWe PumpST; 34,000 MWe TurbineST	0	279.4	15.39	100	74.6	Yes	241.52

(continued on next page)

Table 4 (continued)

Technology Combination	Generation Capacities	Annual Carbon Emission (Mt)	Total Investment (B\$)	Total Annual Cost (B\$)	RES Share in PES (%)	RES in Electricity Produced (%)	CEEP	Annual Electricity Production by RES (TWh/yr)
Generation from four different technologies								
NG, WindOFF, WindON, Hydro	27,400 MW NG; 8,000 MW of WindON; 4,000 MW WindOFF; 10,000 MW of Hydro	68.18	77.6	21.24	16.6	33.2	NO	66.39
NG, WindOFF, WindON, SolarPV	31,000 MW NG; 5,000 MW WindON; 5,000 MW of SolarPV; 7,000 MW WINDOFF	82.36	56.5	23.42	12.9	27.1	NO	46.00
NG, WindOFF, WindON, CSP	30,000 MW NG; 16,000 MW WindON; 6,000 MW CSP; 7,000 MW WINDOFF	62.95	98.3	24.63	19.9	38.3	YES	76.70
NG, WindOFF, Hydro, SolarPV	25,800 MW NG; 16,000 MW SolarPV; 3,000 MW WindOFF; 12,000 MW Hydro	55.31	85.3	18.4	25.3	46	YES	91.93
NG, WindOFF, Hydro, CSP	25,800 MW NG; 4,000 MW CSP; 8,000 MW WindOFF; 13,000 MW of Hydro	59.77	112	23.29	22.1	41.4	NO	82.88
NG, WindOFF, SolarPV, CSP	28,000 MW NG; 5,000 MW CSP; 10,000 MW WindOFF; 45,000 MW SolarPV	43.84	119	20.56	41.2	75.1	YES	150.30
NG, WindON, Hydro, SolarPV	25,000 MW NG; 19,000 MW SolarPV; 10,000 MW WindON; 13,000 MW of Hydro	43.8	930	16.23	35.2	58.2	YES	116.33
NG, WindON, Hydro, CSP	25,800 MW NG; 12,000 MW CSP; 5,000 MW WindON; 13,000 MW Hydro	57.14	145	28.1	23.9	44	YES	88.05
NG, Hydro, SolarPV, CSP	24,000 MW NG; 5,000 MW CSP; 45,000 MW SolarPV; 13,000 MW of Hydro	36.98	133	19.08	47.9	83.1	YES	166.27
NG, WindON, SolarPV, CSP	30,500 MW NG; 9,000 MW CSP; 40,000 MW WindON; 35,000 MW SolarPV	32.34	152	22.28	54.6	95.3	YES	190.68
WindOFF, SolarPV, CSP, ST	44,000 MW WindOFF; 5,000 MW CSP; 45,000 MW SolarPV; 10,000GWh ST; 21,000 MWe PumpST; 30,000 MWe TurbineST	0	289.6	21.37	100	100.6	Yes	243.02
WindOFF, WindON, Hydro, ST	60,000 MW WindON; 10,000 MW WindOFF; 13,000 MW Hydro; 20,000GWh ST; 30,000MWe PumpST; 26,000MWe TurbineST	0	311	19.65	100	83.5	Yes	214.30
WindOFF, Hydro, SolarPV, ST	23,000 MW WINDOFF; 60,000 MW SolarPV; 13,000 MW Hydro; 10,000GWh ST; 15,000MWe PumpST; 21,000MWe TurbineST	0	245	15.29	100	100.3	Yes	243.16
WindOFF, WindON, SolarPV, ST	43,000 MW WindON; 5,000 MW WindOFF; 49,000 MW SolarPV; 12,000GWh ST; 31,000MWe PumpST; 29,000 MWe TurbineST	0	221	13.71	100	83.2	Yes	220.67
WindOFF, Hydro, CSP, ST	36,000 MW WindOFF; 36,000 MW CSP; 13,000 MW Hydro; 8,000 GWh ST; 8,000 MWe PumpST; 21,000 MWe TurbineST	0	423.1	44.36	100	103.72	Yes	233.72
Hydro, SolarPV, CSP, ST	10,000 MW CSP; 80,000 MW SolarPV; 13,000 MW of Hydro; 21,000 GWh ST; 33,000 MWe PumpST; 23,000 MWe TurbineST	0	359.4	24.82	100	88.9	Yes	247.04
WindOFF, WindON, CSP, ST	60,000 MW WindON; 15,000 MW WindOFF; 14,000 MW CSP; 12,000 GWh ST; 34,000 MWe PumpST; 27,000 MWe TurbineST	0	307	26.78	100	82.7	Yes	213.78
WindON, SolarPV, CSP, ST	59,000 MW WindON; 60,000 MW SolarPV; 7,000 MW CSP; 5,000 GWh ST; 30,000 MWe PumpST; 34,000 MWe TurbineST	0	226	17.61	100	107.6	Yes	286.79
WindON, Hydro, SolarPV, ST	36,000 MW WindON; 50,000 MW of SolarPV; 13,000 MW of Hydro; 9,000 GWh ST; 35,000 MWe PumpST; 36,000 MWe TurbineST	0	224	13.44	100	95.2	Yes	242.85
WindON, Hydro, CSP, ST	56,000 MW WindON; 39,000 MW CSP; 13,000 MW of Hydro; 5,000 GWh ST; 10,000 MWe PumpST; 23,000 MWe TurbineST	0	390	42.91	100	112.6	Yes	270.45
Generation from five different technologies								
NG, WindOFF, WindON, Hydro, SolarPV	23,000 MW NG; 21,000 MW WindON; 4,000 MW of WindOFF; 12,000 MW Hydro; 33,000 MW SolarPV	24.91	201.2	12.89	59.9	91.3	Yes	182.62
NG, WindOFF, WindON, Hydro, CSP	24,000 MW NG; 29,000 MW WindON; 4,000 MW WindOFF; 13,000 MW Hydro; 10,000 MW CSP	31.98	164.9	23.34	49.1	74.9	Yes	149.86

(continued on next page)

Table 4 (continued)

Technology Combination	Generation Capacities	Annual Carbon Emission (Mt)	Total Investment (B\$)	Total Annual Cost (B\$)	RES Share in PES (%)	RES in Electricity Produced (%)	CEEP	Annual Electricity Production by RES (TWh/yr)
NG, WindOFF, WindON, SolarPV, CSP	30,000 MW NG; 7,000 MW WindON; 11,000 MW WindOFF; 3,000 MW of SolarPV; 1,000 MW CSP	76.53	71.57	23.77	22.3	42.0	NO	55.74
NG, WindOFF, Hydro, SolarPV, CSP	24,000 MW NG; 5,000 MW of WindOFF; 5,000 MW CSP; 29,000 MW SolarPV; 13,000 MW Hydro	38.29	132.3	19.73	42.6	69.6	Yes	139.26
NG, WindON, Hydro, SolarPV, CSP	25,000 MW NG; 10,000 MW WindON; 4,000 MW of CSP; 18,000 MW SolarPV; 13,000 MW Hydro	42.64	116.1	19.27	36.5	60	Yes	119.95
WindOFF, WindON, Hydro, SolarPV, ST	40,000 MW SolarPV; 10,000 MW WindOFF; 27,000 MW WindON; 13,000 MW Hydro; 11,000 GWh ST; 24,000 MWe PumpST; 23,000 MWe TurbineST	0	237.7	14.70	100	91.9	Yes	227.44
WindOFF, WindON, Hydro, CSP, ST	18,000 MW CSP; 5,000 MW WindOFF; 50,000 MW WindON; 13,000 MW Hydro; 12,000 GWh ST; 18,000 MWe PumpST; 23,000 MWe TurbineST	0	327.3	29.03	100	89.3	Yes	220.3
WindOFF, WindON, SolarPV, CSP, ST	42,000 MW WindON; 10,000 MW WindOFF; 5,000 MW CSP; 37,000 MW SolarPV; 10,500 GWh ST; 29,000 MWe PumpST; 28,000 MWe TurbineST	0	239.5	17.61	100	83.7	Yes	217.88
WindOFF, Hydro, SolarPV, CSP, ST	18,000 MW WindOFF; 50,000 MW SolarPV; 7,000 MW CSP; 13,000 MW Hydro; 9,000 GWh ST; 14,000 MWe PumpST; 20,000 MWe TurbineST	0	257.4	19.2	100	101.1	Yes	248.95
WindON, Hydro, SolarPV, CSP, ST	24,000 MW WindON; 48,000 MW SolarPV; 7,000 MW CSP; 13,000 MW Hydro; 9,000 GWh ST; 14,000 MWe PumpST; 23,000 MWe TurbineST	0	237.8	17.4	100	100	Yes	249.2
Generation from six/all different technologies								
NG, WindOFF, WindON, Hydro, SolarPV, CSP	24,000 MW NG; 23,000 MW SolarPV; 17,000 MW WindON; 1,000 MW WindOFF; 13,000 MW Hydro; 5,000 MW CSP	31.76	134.6	18.56	49.5	76.1	Yes	152.3
WindOFF, WindON, Hydro, SolarPV, CSP, ST	7,000 MW WindOFF; 35,000 MW WindON; 6,000 MW CSP; 35,000 MW SolarPV; 13,000 MW Hydro; 9,000 GWh ST; 25,000 MWe PumpST; 24,000 MWe TurbineST	0	256.4	18.77	100	98.7	Yes	242.82
NG, WindOFF, WindON, Hydro, SolarPV, ST	24,000 MW NG; 20,000 MW SolarPV; 18,000 MW WindON; 4,000 MW WindOFF; 13,000 MW Hydro; 2,000 GWh ST; 6,000 MWe PumpST; 2,000 MWe TurbineST	31.6	127.1	15.82	47.6	68.4	Yes	140.70
NG, WindOFF, WindON, Hydro, SolarPV, CSP, ST	18,000 MW NG; 26,000 MW SolarPV; 37,000 MW WindON; 30,000 MW WindOFF; 13,000 MW Hydro; 5,000 MW CSP; 3,000 GWh ST; 4,000 MWe PumpST; 5,000 MWe TurbineST	4.67	249.6	21.36	92.6	128.4	Yes	286.36

CSP: Concentrated solar power plant, **NG:** Natural gas power plant, **Hydro:** River hydropower plant, **SolarPV:** Solar photovoltaic power plant, **PumpST:** Hydro-storage system pump capacity, **ST:** Pumped hydro-storage system, **TurbineST:** Hydro-storage system turbine capacity, **WindOFF:** Offshore wind power plant, **WindON:** Onshore wind power plant.

total annual cost of scenarios with zero yearly CO₂ emission are significantly higher than NG integrated scenarios. The use of NG, WindOFF, WindON, and SolarPV power plants to meet the estimated grid electricity demand is the most cost-effective option (with an investment cost of \$ 76.9B and an annual total cost of \$ 17.27B). This will require 29,000 MW of NG, 7000 MW of WindOFF, 25,000 MW of SolarPV, and 9000 MW of WindON. Considering the existing power system in the country, the use of NG, WindOFF, WindON, and Hydro powerplants will be the most preferable option as NG and Hydro powerplants are already in use in the country. This will reduce investment cost by a significant amount as the current economic analysis in this study is based on the installation of all the systems newly. In comparison to other combinations for this category, this scenario (NG, WindOFF, WindON, and Hydro) is one of the two scenarios with no CEEP. The most (environmentally) sustainable plan is the combination of pumped hydro storage

(ST) with WindON, CSP, and SolarPV. It requires the least economic commitments out of all the zero CO₂ emitting scenarios. In the combination of four technologies, there are different pros and cons for each of the scenarios, however, there are some combinations that seem unattainable due to the excessive investment and yearly cost. These scenarios are WindON, Hydro, CSP, and ST; WindOFF, Hydro, CSP, and ST; and Hydro, SolarPV, CSP, and ST. The high cost of these scenarios can be attributed to the installation (invest) and running of CSP, WindOFF, and ST.

In comparison to the use of three or four technologies, the average cost for all the scenarios with a probable solution with five technologies is higher. While the cheapest option is the combination of NG, WindOFF, WindON, CSP, and SolarPV, the difference in cost when considering the integration of Hydro technology is significant. This indicates that the country will have to neglect its hydropower resource if they are to

prioritize cost. However, in a case where the country is trying to achieve an emission-free power sector, the integration of Hydro energy resources will be utilized as seen in Table 4. Another outstanding observation that applies to all the scenarios that integrate five different technologies is that this will leave CEEP in the PES. This applies to the combination of six technologies as well as the use of all the technologies. Therefore, the integration of hydrogen production and BEVs will be paramount to the maximization of the electricity produced in these scenarios. One obvious fact is that the total investment cost difference (over \$ 122.2B) between zero-emission scenarios and NG powerplant integrating scenarios for six technological combinations is significantly huge. However, the total annual costs are comparable (Table 4). The use of all the technologies will also have CEEP in the PES, although this can be eradicated with the increase in the share of NG power plant integration. This scenario and some other scenarios from Table 4 have been further analyzed by integrating BEVs and Hydrogen production and the results are discussed in the subsequent section.

3.2. Viability of integrating BEVs and hydrogen production with the feasible solutions for different combinations

The results from the preceding section highlight the optimal capacities required from different technologies in order to meet the estimated electricity demand. Priority has been given to RES-based and low-cost technologies. Also, each scenario is simulated to have low or no CEEP in the power system network. Nevertheless, a scenario is only considered a feasible solution if and only if it can meet all the electricity demand for every hour base on the load profile used (i.e., no PP import error as stated in Section 2). The integration of BEVs and hydrogen production is solely to eradicate the projected CEEP within this power system network. In this section, different scenarios for different types of technological combinations are presented while highlighting the importance of BEVs and hydrogen production. Since the use of the NG power plant is the only feasible single technology, this is neglected for BEVs and hydrogen production as there will be no CEEP with the use of this technology. Furthermore, for dual technological combinations, the integration NG with other RES leaves no CEEP in the power system network as seen in Table 4. Therefore, these scenarios are not considered for further analysis but two different scenarios (ST with either SolarPV or WindON) have been chosen for further analysis. In general, the integration of NG power plants is not considered for BEVs and hydrogen

production integration as this has no CEEP for most of the scenarios. Also, the aim of integrating hydrogen production and BEVs in this study is to enhance RES integrations. The results are plotted and illustrated in Figs. 4–14. The y-axis of these figures shows the hourly electricity demand/supply in MW while the x-axis represents the hours throughout the year. Different colors have been used to represent the electricity production profile of the considered RE-based technologies. The figures (Figs. 4–14) are plotted based on the hourly data from the results of the analysis in this study.

The required capacities for SolarPV and WindON are 90,000 MW and 100,000 MW and then this will be integrated with ST considering varying capacities as seen in Table 4. The total annual electricity production from RES is 241.52 TWh/yr for SolarPV and 231.19 TWh/yr for WindON. Considering the estimated yearly electricity demand capacity, the excess electricity from the two scenarios are 41.52 TWh/yr and 31.19 TWh/yr for SolarPV and WindON respectively. However, some of the excess electricity (41.52 TWh/yr from SolarPV; 31.19 TWh/yr from WindON) produced will be used in the process of energy conversion by the pumped hydro storage system integrated with this SolarPV and WindON for dual technology scenarios. In this study, the BEVs integrated with the energy plan are expected to charge for 12 h during the night/morning hours as this is the average timeframe and period BEVs are charged in countries like China that are actively utilizing such.

For WindON and ST scenarios, the maximum BEVs charge demand that can be integrated into the power system network is 0.03 TWh/yr and this will allocate 7 MW of electricity for BEV charging for 12 h. The electricity demand and supply profile for this scenario is presented in Fig. 4. The remaining CEEP from the system is 15.02 TWh/yr and this will produce a total of 2.85×10^8 kg of hydrogen annually. The production profile for the hydrogen generated is presented in the first chart in Fig. 5. For SolarPV and ST scenarios the maximum BEV demand that can be integrated with the electricity network is 0.34 TWh/yr and this corresponds to 77 MW of electricity consumption by BEVs during the charging period. The electricity demand and supply profile (Fig. 6) for this case scenario (SolarPV and ST) show that the excess electricity will be throughout the year while the ST system will subsidize the lack of electricity production by the SolarPV power plant during night hours. In order to meet the hourly electricity demand, the required SolarPV capacity is 90,000 MW and the ST system will have a storage of 3000 GWh with a pump and turbine capacity of 51,000 MW and 36,000 MW respectively. The hydrogen produced from the excess electricity

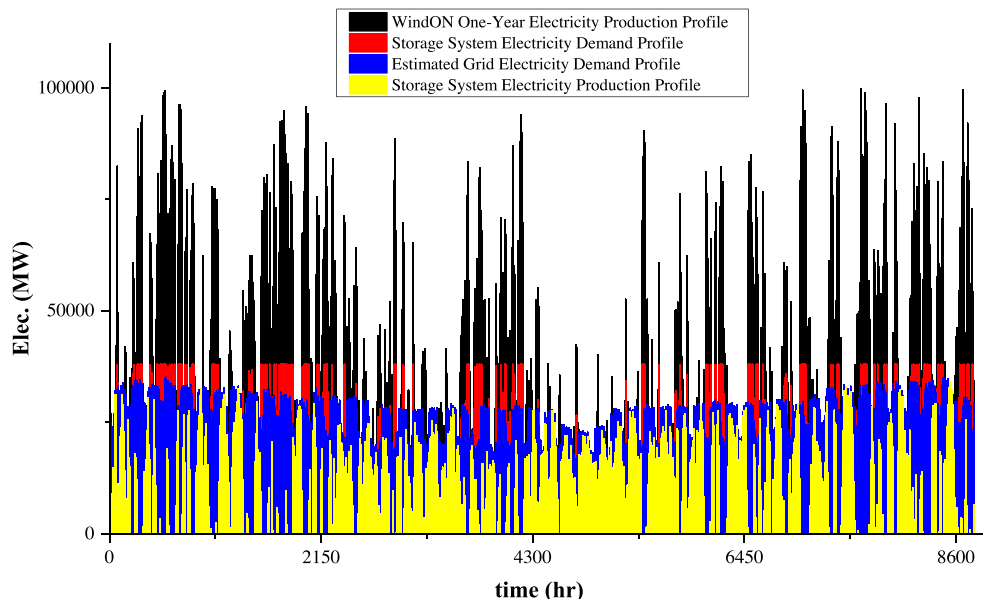


Fig. 4. WindON and ST annual electricity demand and supply profile.

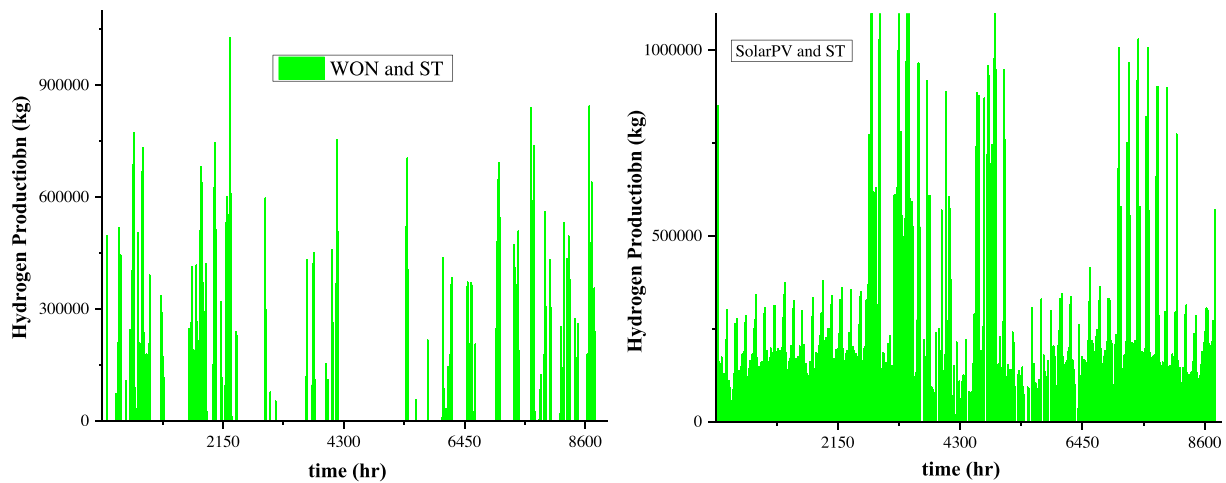


Fig. 5. WindON/ST and SolarPV/ST annual Hydrogen production profile.

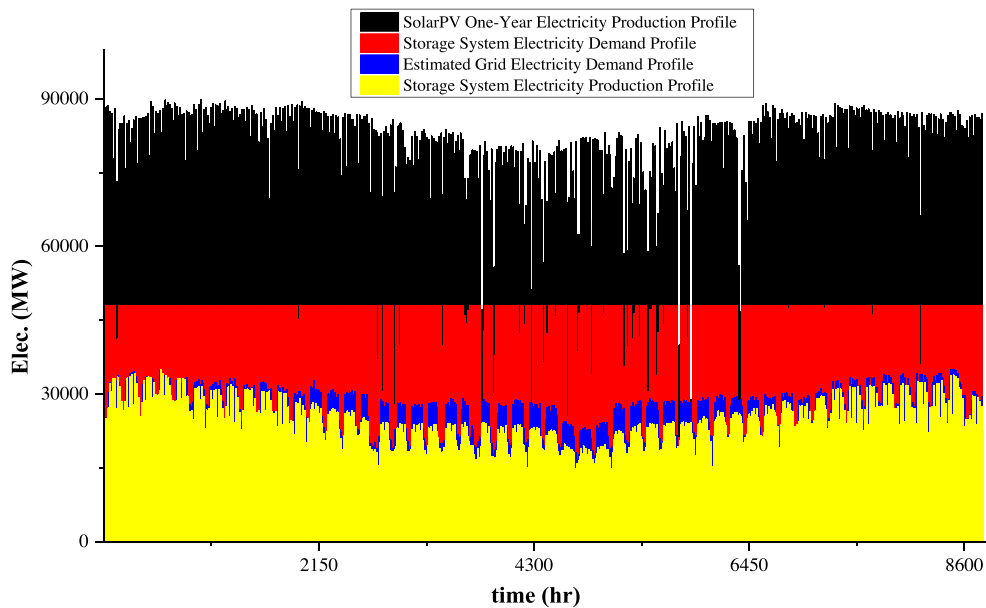


Fig. 6. SolarPV and ST annual electricity demand and supply profile.

production in this scenario (SolarPV and ST) is a reflection of the CEEP in the power system network. As seen in Fig. 5 over 3.37×10^8 kg of hydrogen will be produced during some hours and this will require a huge installation of the hydrogen generation plant. Considering the economic commitment and the technicality of dual technology electricity generation from RES only, the use of SolarPV is more preferable. Also, as earlier stated in Section 2, the use of WindON for commercial or private electricity generation in Nigeria is very insignificant in comparison to the adoption of SolarPV.

From the economic figures presented in Table. 2, SolarPV and WindON are the two cheapest technologies. The integration of hydrogen production and BEVs will be considered for WindON, SolarPV, and ST in the tri-technological combination scenarios as this is the most economically feasible for RES-based scenarios. The required installation capacities to meet the electricity demand are 60,000 MW for WindON, 75,000 MW for SolarPV, and 4000 GWh for ST with 29,000 MWe PumpST and 34,000 MWe TurbineST (the electricity demand and supply profile for this case are plotted in Fig. 7). The total hydrogen production from the 92.78 TWh/yr CEEP is 1.76×10^9 kg and the BEVs demand that can be integrated with the power system network is 1.11 TWh/yr. This

amounts to a 253 MW electricity consumption by the BEVs during the charging hours. The huge hydrogen production recorded for this case scenario is a reflection of the large integration of SolarPV and WindON which directly impacts the CEEP in the power system network.

Furthermore, the addition of Hydro technology to WindON, SolarPV, and ST is the cheapest option for electricity with four RES technology only. The optimal capacities are 36,000 MW of WindON, 13,000 MW of Hydro, and 50,000 MW of SolarPV as well as 5000 GWh of ST (Table. 4). This will also produce zero carbon emission and the total investment cost is estimated to be \$ 224B. The electricity demand and supply profile for this scenario (Fig. 7) show that all the technologies are actively involved in the daily generation of electricity. The maximum electricity consumption by the BEVs integrated with the power network for this scenario is 0.03 TWh/yr and this results in a 7 MW electricity consumption by the BEVs during the charging hours. Also, the remaining CEEP (17.73 TWh/yr) for this case scenario is smaller in comparison to other cases. A total of 6.15×10^8 kg of hydrogen can be produced from the and the maximum hydrogen generation in an hour is 1,272,620 kg (Fig. 8). The electricity demand and supply profile over the period of one year (Fig. 9) show the active participation of the ST system in electricity

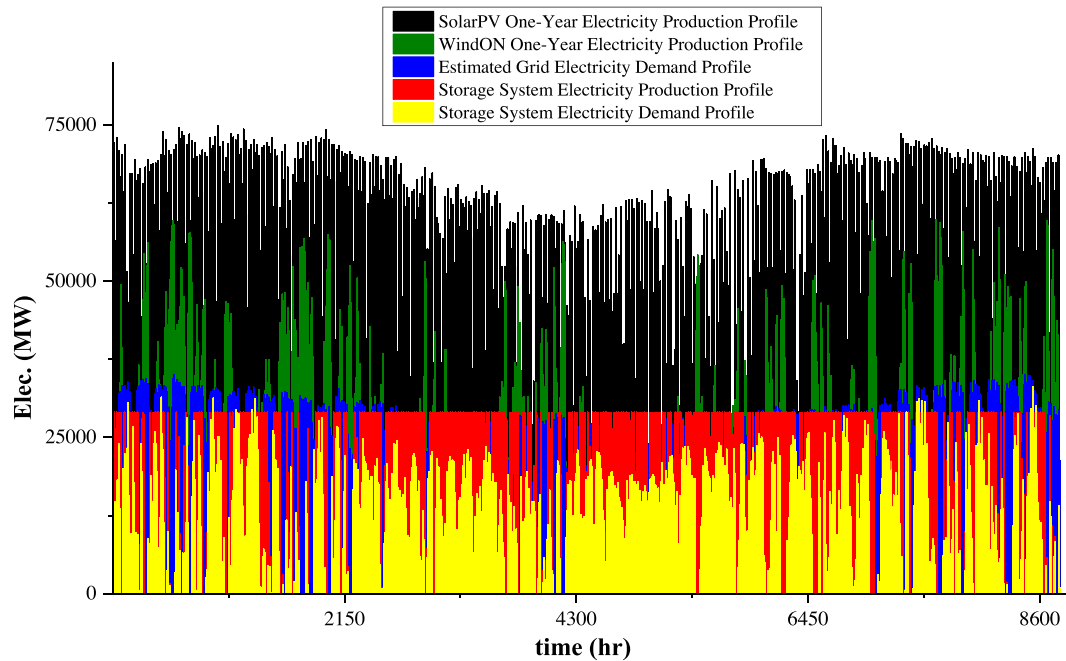


Fig. 7. SolarPV, WindON, and ST annual electricity demand and supply profile.

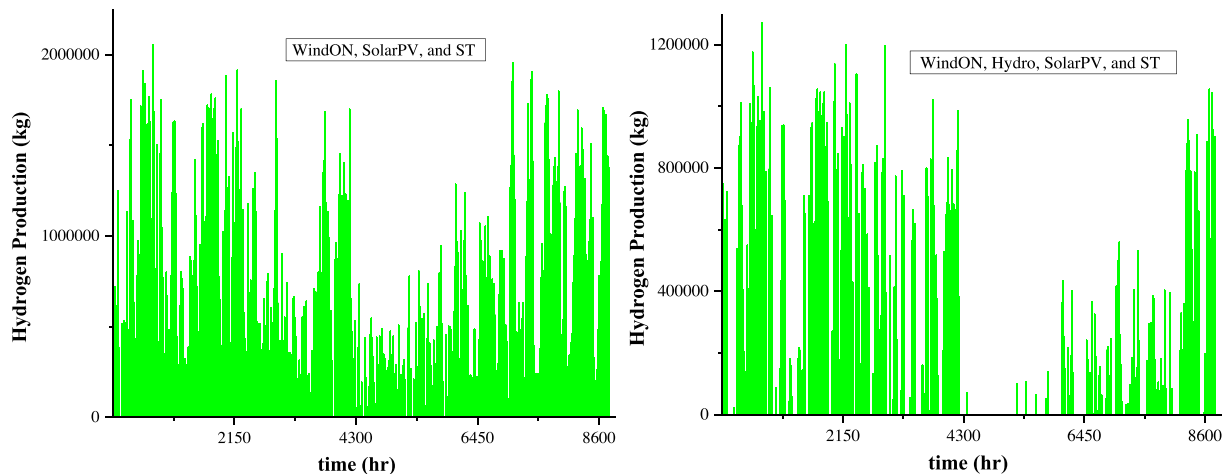


Fig. 8. Hydrogen production profile for tri and quad-technological combinations.

stability.

In modern-day energy planning, beyond the generation of enough electricity to meet the needs of a country, the environmental sustainability and energy security of the electricity generation source are also crucial. Most developed countries have adopted the use of different energy sources and technologies for electricity generation in order to improve their energy security/diversity. The generation of electricity from five or more of the technologies considered in the scope of this study provides an opportunity for Nigeria to compete with developed countries in terms of electricity security and energy diversity. The use of NG, WindOFF, WindON, SolarPV, and CSP is the most economic option as seen in Table 4. However, the integration of hydrogen production and BEVs is only analyzed for the most economically viable RES-based combination. This combination involves the use of 40,000 MW of SolarPV, 10,000 MW of WindOFF, 27,000 MW of WindON, 13,000 MW of Hydro, as well as 11,000 GWh of ST with 23,000 MWe TurbineST and 24,000 MWe of PumpST. Fig. 10 illustrates the involvement of all these technologies in the annual electricity supply and it is obvious that the power system is more secured with the integration and participation of

these technologies in electricity generation. As highlighted in Table 4, the RES share in electricity production is 91.9% while the remaining share is augmented by the pumped hydro storage system (ST). Although the total annual cost (\$ 14.7B) for this case scenario is also competitive with the other NG integrated scenarios, the total investment cost (\$ 237.7B) is relatively higher. This technological combination can accommodate 0.094 TWh/yr of BEVs load based on this load profile used in this study and the hourly charging demand of 11 MW is totally catered for. The implementation of this case scenario will also lead to the production of an enormous amount of hydrogen from the CEEP (as seen in Fig. 11).

Also, the utilization of all the RE-based technologies considered in the scope of this research can further strengthen the energy diversity and environmental sustainability of Nigeria. The consistent solar radiation throughout the year makes the integration of SolarPV and CSP paramount to the production of emission-free electricity for the country. The capacities of each respectively are 40,000 MW for SolarPV, 13,000 MW for Hydro, 27,000 MW for WindON, 10,000 MW for WindOFF, and 11,000 GWh for ST. This will require a total investment of \$ 237.7B and

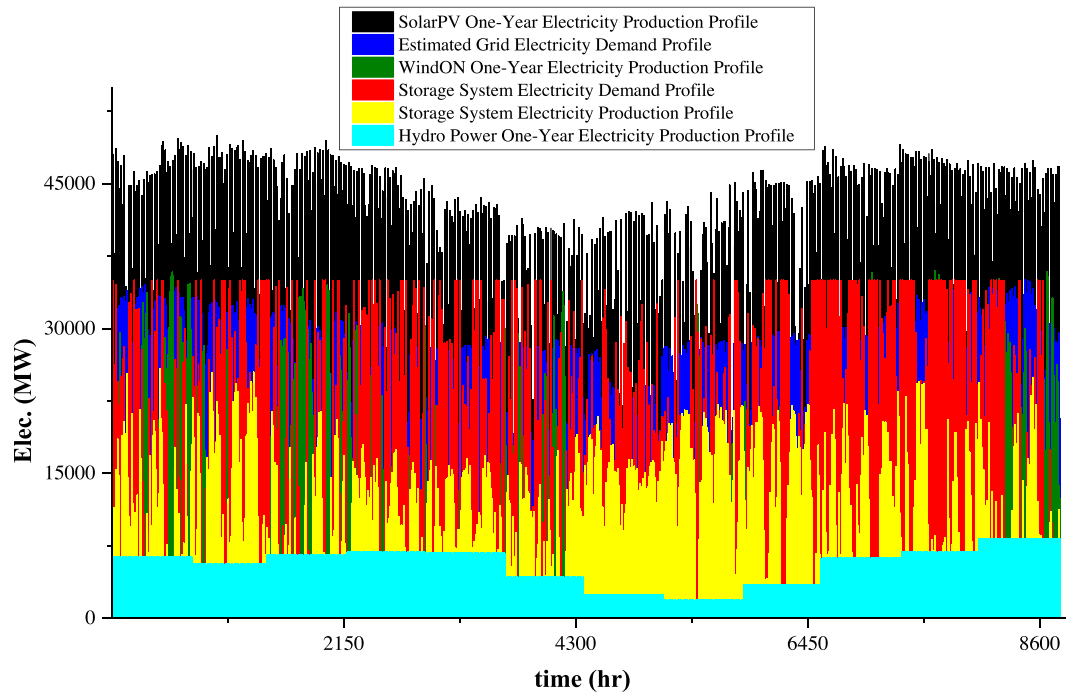


Fig. 9. SolarPV, CSP, WindON, and ST annual electricity demand and supply profile.

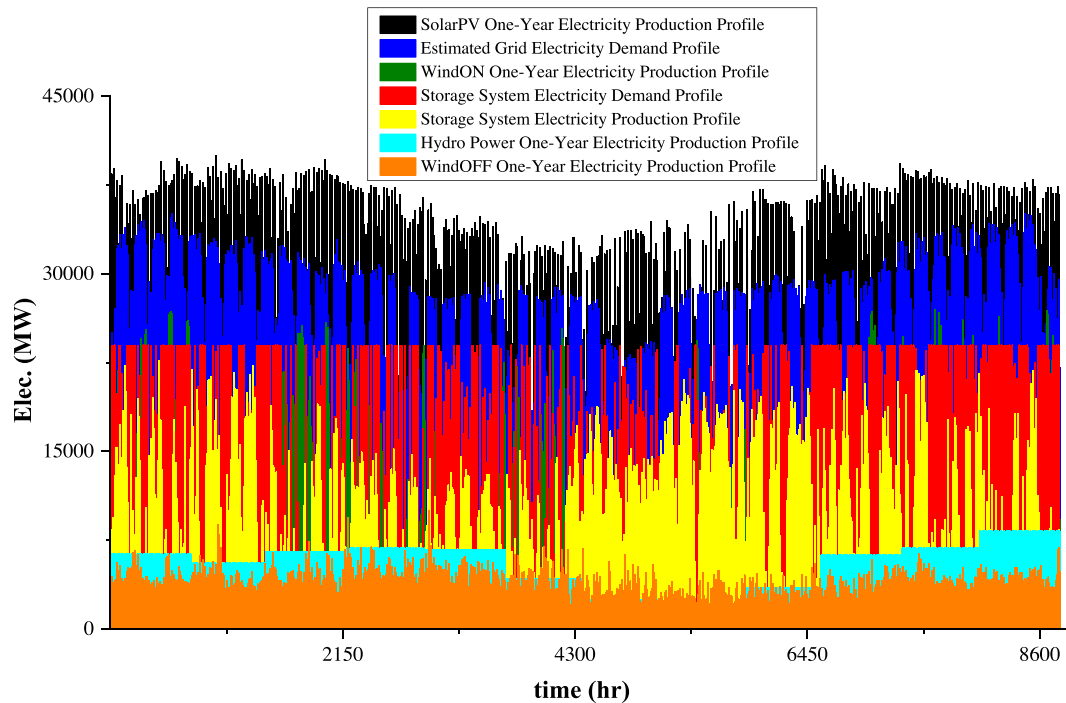


Fig. 10. SolarPV, Hydro, WindON, WindOFF, and ST annual electricity demand and supply profile.

an annual investment cost of \$ 14.70B. The total electricity production is 227.44 TWh/yr and 91.9% of this is produced by RES. The annual total and hourly maximum hydrogen production from the CEEP in this case scenario are 6.48×10^8 kg and 1,148,210 kg respectively (Fig. 11). The electricity demand and supply profile (Fig. 12) also reflects the intermittent and stochastic nature of the different RES. The addition of NG technology to all the RES technologies is also analyzed as this will highlight a scenario where all the technologies in the study are combined. It is noteworthy that the addition of NG reduced the total

investment cost of \$ 256.4B for all RES technologies (Fig. 12) to \$ 249.6B as seen in Table 4. However, the annual investment cost increases by \$ 2.59B yearly. The increment in annual investment cost is due to the additional price of natural gas consumed by the power plant. Also, the capacity of NG required is 18,000 MW and this will emit 4.67 Mt of CO₂ yearly. The share of RES in the PES is 92.6% while the electricity produced by the RES technologies is 128.4% of the required. The total electricity produced over a period of one year is 286.36 TWh/yr (Table. 4). The high CEEP in the power system network is a reflection of

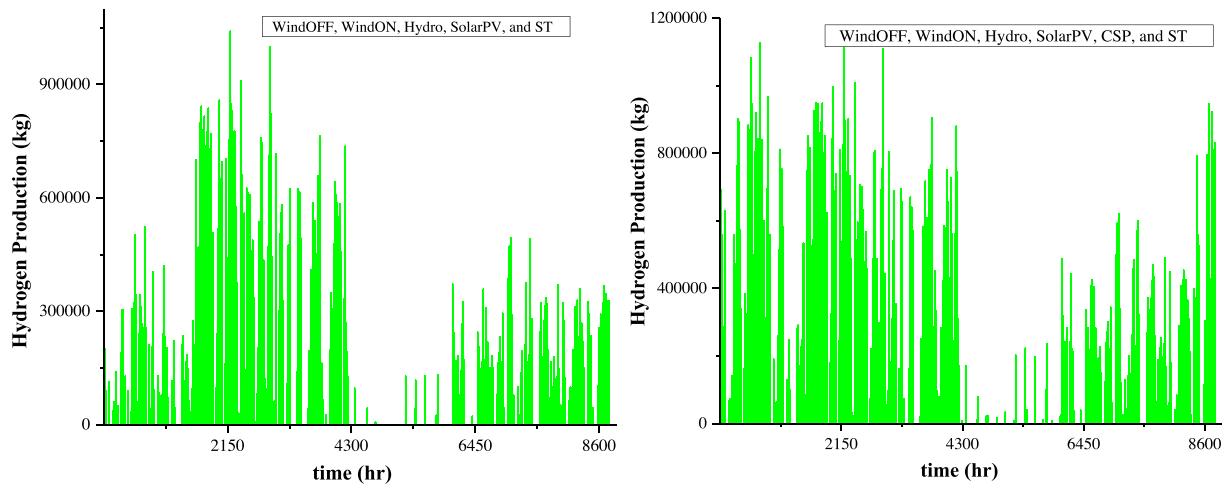


Fig. 11. Hydrogen production profile for five and six technological combinations.

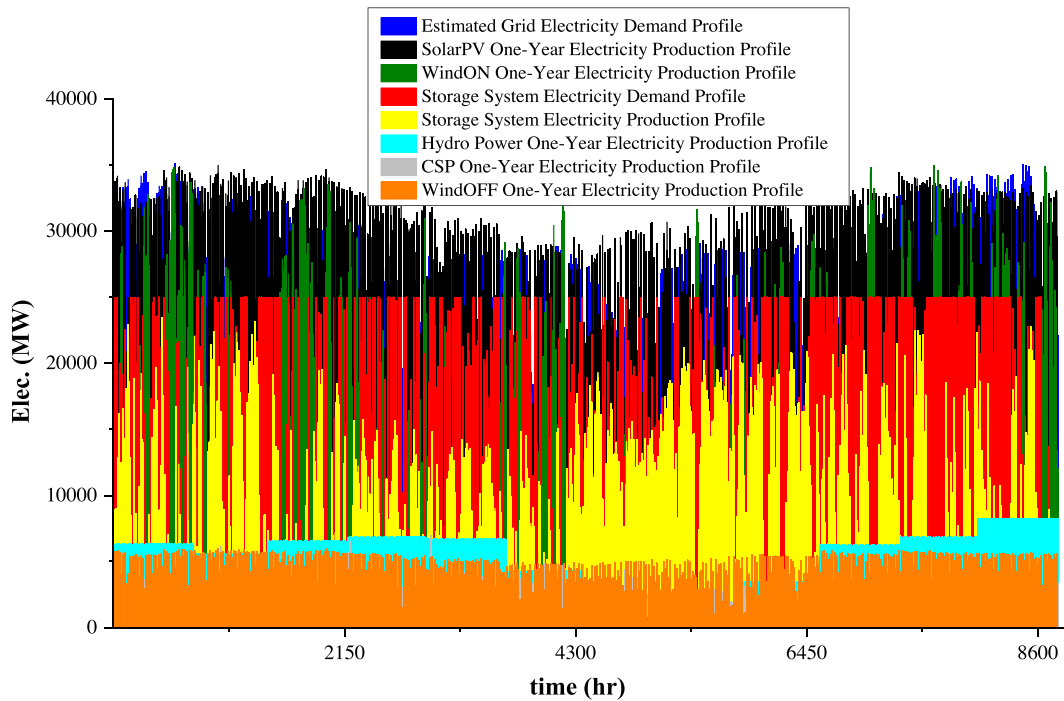


Fig. 12. SolarPV, Hydro, WindON, CSP, WindOFF, and ST annual electricity demand and supply profile.

the hydrogen production plotted in Fig. 14. It is also noteworthy that the excess electricity for this case scenarios allowed the highest BEV integration (8.66 TWh/yr) and the electricity consumable by BEVs during charging hours is 986 MW (as seen in Fig. 13).

4. Conclusions

In this study, a comprehensive approach that can be used to solve the imminent electrification challenges in developing countries is presented considering the generation side of electricity production. EnergyPLAN and Matlab computer programs have been used to implement the proposed/developed method. The analysis presented in this study is done on an hourly time-step and the solutions presented are the possible scenarios with which the hourly electricity demand can be satisfied. The use of seven technologies namely; NG, WindOFF, WindON, SolarPV, CSP, Hydro, and ST have been considered in this study and the data used have been extracted from reliable sources. Although Nigeria is

considered as the case area for the implementation of the method proposed, this method can be used for other developing countries with a similar challenge. Beyond providing feasible scenarios and plans to tackle the existing electricity production in Nigeria, guidelines on steps (not to take) to tackle this problem are embedded in this study. With the analysis of the seven aforementioned technologies (considering different combinations), the viable plans considering cost and CO₂ emission are determined from this study. The concluding points are summarized below.

- Based on the analysis, the use of NG is the only feasible singular technology that can be used to meet the electricity demand in order to achieve a stable electricity supply. One of the main drawbacks of the power sector is the inadequate supply of natural gas in many developing countries. Although the initial investment of NG as the sole producer of electricity is the lowest in comparison to other

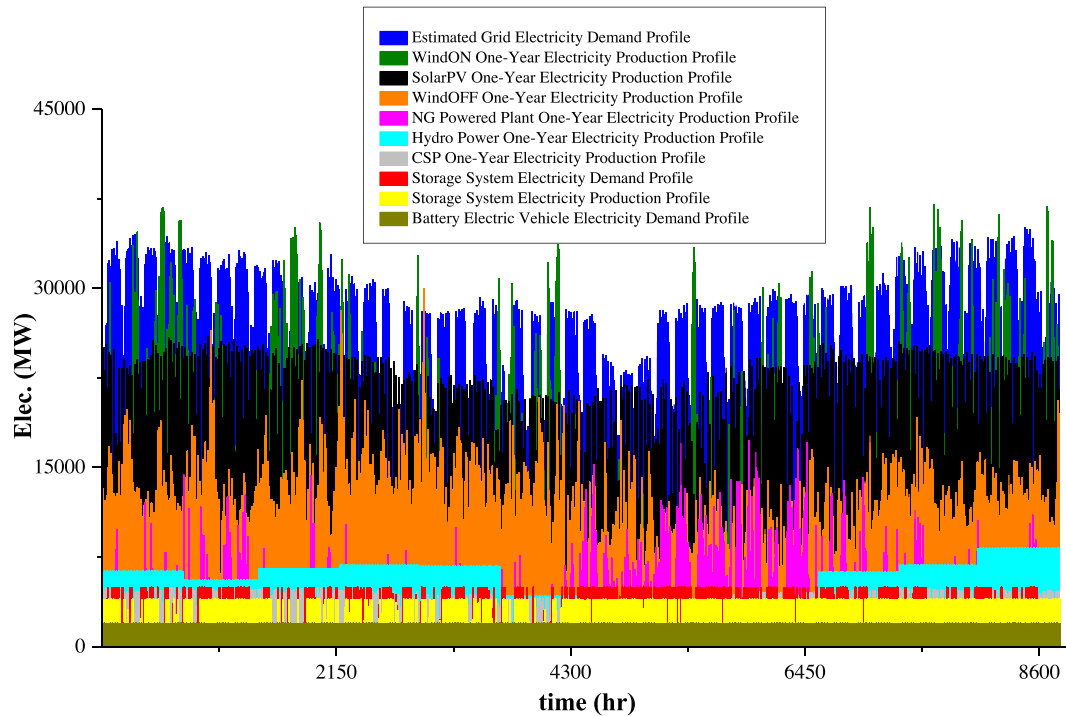


Fig. 13. NG, SolarPV, Hydro, WiindON, CSP, WindOFF, and ST annual electricity demand and supply profile.

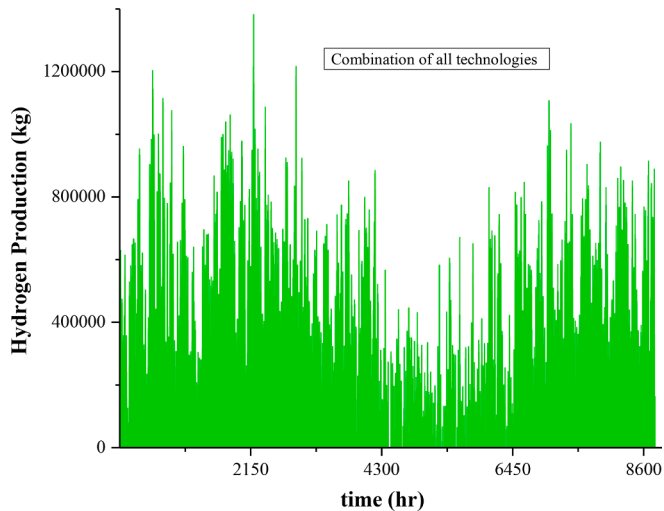


Fig. 14. Hydrogen production profile for all technological combinations.

scenarios in this study, the carbon emission and social instability attached to this technology make the solution less feasible.

- The intermittent nature of the RE-based technologies considered in the study and the insufficient hydro potential in Nigeria makes the use of singular RE technologies impossible. The most cost-effective solution with the use of RES and non-RES is the combination of NG, WindON, and SolarPV. This requires 32,500 MW of NG, 12,000 MW of SolarPV, and 7000 MW of WindON while the total investment cost and total annual cost are \$ 48.7B and \$ 22B respectively.
- For all the scenarios and technological combinations considered in this study, the cost of using RE-based technology is twice (or more) the cost of integrating RES and non-RES. Therefore, it can be concluded that fossil fuels are integral to the attainment of an economically feasible pathway to the 100% electrification of many developing countries. The most viable solution to the poised problem

based on RES only is the combination of 36,000 MW of WindON, 13,000 MW of Hydro, 50,000 MW of SolarPV, and 9000 GWh of ST (with 35,000 MWe PumpST and 36,000 MWe of TurbineST). This is also the most feasible pathway to achieving emission-free power generation, however, it requires a high total investment (\$ 224B).

- Contrary to some results in existing studies that reported the integration of storage systems as a means of eradicating electricity wastage from the PES, based on the results of this study, it can be concluded that the integration of ST will reduce electricity waste (CEEP) for most of the scenarios. Although CEEP is eradicated with ST for some scenarios, the integration of ST that can eradicate CEEP (in other scenarios) will increase the economic values making it unattainable.
- The integration of plug-in BEVs and hydrogen production will help maximize the electricity produced from the RE-based power systems. This will also reduce the overall carbon emission from the country at large. Considering the most viable RE technological combinations (Hydro, WindON, SolarPV, and ST), about 6.15×10^8 kg of hydrogen is producible from the CEEP for this scenario. Also, the plug-in BEVs integrated with this scenario will consume 7 MW of electricity during the charging hours and this will use a total of 0.03 TWh/yr from the CEEP.
- In order to meet the Nigerian federal government targets of over 30% integration of renewables into the energy (electricity generation) mix by 2030, the use of NG and Hydro or the use of NG, WindOFF, WindON, SolarPV, and CSP will be required. The capacities, total investment costs, and other parameters for these scenarios are presented in Table. 4.
- The method developed in this study will be helpful for industrial application and technical analysis as it can be adopted by countries with a similar electrification challenge. Also, beyond the RES considered within the scope of this study, the integration of other RE-based technologies can be used in the proposed method.
- Finally, this study can be used as a template for the stepwise huge application and integration of RES technologies for electricity. It will also help determine the sustainability of a chosen pathway with/without the integration of EVs and hydrogen production.

In future studies, the role of the transmission/distribution network in the 100% electrification proposal will be analyzed in detail. Also, the method proposed will be of great value to industries saddled with the task to analyse and install national-scale RE-based power systems.

CRedit authorship contribution statement

Olusola Bamisile: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing. **Akinola Babatunde:** Writing - original draft, Investigation, Methodology. **Humphrey Adun:** Writing - original draft, Investigation, Methodology. **Nasser Yimen:** Methodology, Writing - review & editing, Validation. **Mustapha Mukhtar:** Investigation, Resources, Validation. **Qi Huang:** Supervision, Funding acquisition, Validation. **Weihao Hu:** Supervision, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Dioha M. Nigeria's renewable energy policy: a fantasy or reality? *Renewable Energy World* 2018.
- [2] IRENA. *Renewable Energy Country Rankings*. International Renewable Energy Agency; 2018.
- [3] Zhiznin SZ, Timokhov VM, Gusev AL. In: Economic aspects of nuclear and hydrogen energy in the world and Russia; 2020. <https://doi.org/10.1016/j.ijhydene.2020.08.260>.
- [4] Martin A, Agnoletti MF, Brangier E. Users in the design of Hydrogen Energy Systems: a systematic review. *Int J Hydrogen Energy* 2020;45(21):11889–900. <https://doi.org/10.1016/j.ijhydene.2020.02.163>.
- [5] Liu Z, Yu L. Stay or leave? The role of air pollution in urban migration choices. *Ecol Econ* 2020;177:106780. <https://doi.org/10.1016/j.ecolecon.2020.106780>.
- [6] IEA. *The Future of Hydrogen – Analysis*. IEA Publications; 2019.
- [7] Vivas FJ, Segura F, Andújar JM, Caparrós JJ. A suitable state-space model for renewable source-based microgrids with hydrogen as backup for the design of energy management systems. *Energy Convers Manag* 2020;219:113053. <https://doi.org/10.1016/j.enconman.2020.113053>.
- [8] Sezer N, Koç M. Development and performance assessment of a new integrated solar, wind, and osmotic power system for multigeneration, based on thermodynamic principles. *Energy Convers Manag* 2019;188:94–111. <https://doi.org/10.1016/j.enconman.2019.03.051>.
- [9] Dahlke C, Sterling Steven, Meehan John. Policy and market drivers for advancing clean energy. In: *Advances in Clean Technologies*. Academic Press; 2021. p. 451–85.
- [10] Wang R, Hsu S-C, Zheng S, Chen J-H, Li XI. Renewable energy microgrids: Economic evaluation and decision making for government policies to contribute to affordable and clean energy. *Appl Energy* 2020;274:115287. <https://doi.org/10.1016/j.apenergy.2020.115287>.
- [11] Kotzebue JR, Weissenbacher M. The EU's Clean Energy strategy for islands: a policy perspective on Malta's spatial governance in energy transition. *Energy Policy* 2020;139:111361. <https://doi.org/10.1016/j.enpol.2020.111361>.
- [12] Ghandi A, Paltsev S. Global CO2 impacts of light-duty electric vehicles. *Transp Res Part D Transp Environ* 2020;87:102524. <https://doi.org/10.1016/j.trd.2020.102524>.
- [13] Kobashi T, Say K, Wang J, Yarime M, Wang D, Yoshida T, et al. Techno-economic assessment of photovoltaics plus electric vehicles towards household-sector decarbonization in Kyoto and Shenzhen by the year 2030. *J. Clean. Prod.* 2020; 253:119933. <https://doi.org/10.1016/j.jclepro.2019.119933>.
- [14] IEA. *Global EV Outlook: scaling up the transition to electric mobility – CityLogistics*; 2019.
- [15] Habib S, Kamran M, Rashid U. Impact analysis of vehicle-to-grid technology and charging strategies of electric vehicles on distribution networks - a review. *J Power Sources* 2015;277:205–14. <https://doi.org/10.1016/j.jpowsour.2014.12.020>.
- [16] Steward D. Critical Elements of Vehicle-to-Grid (V2G) Economics; 2017.
- [17] Drude L, Pereira Junior LC, Rütther R. Photovoltaics (PV) and electric vehicle-to-grid (V2G) strategies for peak demand reduction in urban regions in Brazil in a smart grid environment. *Renewable Energy* 2014;68:443–51. <https://doi.org/10.1016/j.renene.2014.01.049>.
- [18] Zhou X, Lukic S, Bhattacharya S, Huang A. Design and control of grid-connected converter in bi-directional battery charger for plug-in hybrid electric vehicle application. In: 5th IEEE Vehicle Power and Propulsion Conference, VPPC '09; 2009. p. 1716–21. <https://doi.org/10.1109/VPPC.2009.5289691>.
- [19] Zhou X, Wang G, Lukic S, Bhattacharya S, Huang A. Multi-function bi-directional battery charger for plug-in hybrid electric vehicle application. In: 2009 IEEE Energy Conversion Congress and Exposition, ECCE 2009; 2009. p. 3930–6. <https://doi.org/10.1109/ECCE.2009.5316226>.
- [20] Colak I, Sagioglu S, Fulli G, Yesilbudak M, Covrig CF. A survey on the critical issues in smart grid technologies. *Renewable Sustainable Energy Rev* 2016;54: 396–405. <https://doi.org/10.1016/j.rser.2015.10.036>.
- [21] Shang DR, Sun G. Electricity-price arbitrage with plug-in hybrid electric vehicle: Gain or loss? *Energy Policy* 2016;95:402–10. <https://doi.org/10.1016/j.enpol.2016.05.019>.
- [22] Kesselring R. The electricity crisis in Zambia: Blackouts and social stratification in new mining towns. *Energy Res Soc Sci* 2017;30:94–102. <https://doi.org/10.1016/j.erss.2017.06.015>.
- [23] Shaikh F, Ji Q, Fan Y. The diagnosis of an electricity crisis and alternative energy development in Pakistan. *Renewable Sustainable Energy Rev* 2015;52:1172–85. <https://doi.org/10.1016/j.rser.2015.08.009>.
- [24] Poudyal R, Loskot P, Nepal R, Parajuli R, Khadka SK. Mitigating the current energy crisis in Nepal with renewable energy sources. *Renewable Sustainable Energy Rev* 2019;116:109388. <https://doi.org/10.1016/j.rser.2019.109388>.
- [25] Doll CNH, Pachauri S. Estimating rural populations without access to electricity in developing countries through night-time light satellite imagery. *Energy Policy* 2010;38(10):5661–70. <https://doi.org/10.1016/j.enpol.2010.05.014>.
- [26] United Nations, Nigeria Population (2020) – Worldometer; 2020.
- [27] Ugwoke B, Gershon O, Becchio C, Corgnati SP, Leone P. A review of Nigerian energy access studies: the story told so far. *Renewable and Sustainable Energy Reviews* 2020;120. <https://doi.org/10.1016/j.rser.2019.109646>.
- [28] Al-Sumaiti AS, Salama MMA, El-Moursi M. Enabling electricity access in developing countries: a probabilistic weather driven house based approach. *Appl Energy* 2017;191:531–48. <https://doi.org/10.1016/j.apenergy.2017.01.075>.
- [29] Fu T, Jian Z. A developmental state: How to allocate electricity efficiently in a developing country. *Energy Policy* 2020;138:111231. <https://doi.org/10.1016/j.enpol.2019.111231>.
- [30] Bos K, Chaplin D, Mamun A. Benefits and challenges of expanding grid electricity in Africa: a review of rigorous evidence on household impacts in developing countries. *Energy Sustainable Dev* 2018;44:64–77. <https://doi.org/10.1016/j.esd.2018.02.007>.
- [31] Dada JO. Towards understanding the benefits and challenges of Smart/Micro-Grid for electricity supply system in Nigeria. *Renewable Sustainable Energy Rev* 2014; 38:1003–14. <https://doi.org/10.1016/j.rser.2014.07.077>.
- [32] Ali HS, Nathaniel SP, Uzuner G, Bekun FV, Sarkodie SA. Trivariate modelling of the nexus between electricity consumption, urbanization and economic growth in Nigeria: fresh insights from Maki Cointegration and causality tests. *Heliyon* 2020;6 (2). <https://doi.org/10.1016/j.heliyon.2020.e03400>.
- [33] Adhekpukoli E. The democratization of electricity in Nigeria. *Electr J* 2018;31(2): 1–6. <https://doi.org/10.1016/j.tej.2018.02.007>.
- [34] Odekanle EL, Odejobi OJ, Dahunsu SO, Akeredolu FA. Potential for cleaner energy recovery and electricity generation from abattoir wastes in Nigeria. *Energy Rep* 2020;6:1262–7. <https://doi.org/10.1016/j.egyr.2020.05.005>.
- [35] Ogunmodimu O, Okoroigwe EC. Solar thermal electricity in Nigeria: prospects and challenges. *Energy Policy* 2019;128:440–8. <https://doi.org/10.1016/j.enpol.2019.01.013>.
- [36] Afful-Dadzie A, Afful-Dadzie E, Abbey NA, Owusu BA, Awudu I. Renewable electricity generation target setting in developing countries: modeling, policy, and analysis. *Energy Sustainable Dev.* 2020;59:83–96. <https://doi.org/10.1016/j.esd.2020.09.003>.
- [37] Oyewo AS, Aghahosseini A, Bogdanov D, Breyer C. Pathways to a fully sustainable electricity supply for Nigeria in the mid-term future. *Energy Convers Manag* 2018; 178(October):44–64. <https://doi.org/10.1016/j.enconman.2018.10.036>.
- [38] Pietrosemoli L, Rodríguez-Monroy C. The Venezuelan energy crisis: Renewable energies in the transition towards sustainability. *Renewable Sustainable Energy Rev* 2019;105:415–26. <https://doi.org/10.1016/j.rser.2019.02.014>.
- [39] Offiong P. Nigerian Renewal Energy and Energy Efficiency Policy (NREEEP) - Climate Scorecard; 2019.
- [40] Østergaard PA. Reviewing EnergyPLAN simulations and performance indicator applications in EnergyPLAN simulations. *Appl Energy* 2015;154:921–33. <https://doi.org/10.1016/j.apenergy.2015.05.086>.
- [41] Fernandes L, Ferreira P. Renewable energy scenarios in the Portuguese electricity system. *Energy* 2014;69:51–7. <https://doi.org/10.1016/j.energy.2014.02.098>.
- [42] Novosel T, Pukšec T, Krajačić G, Duić N. Role of district heating in systems with a high share of renewables: case study for the City of Osijek. *Energy Procedia* 2016; 95:337–43. <https://doi.org/10.1016/j.egypro.2016.09.019>.
- [43] Connolly VMD, Lund H. *Smart Energy Europe: The technical and economic impact of one potential 100% renewable energy scenario for the European Union*. *Renewable Sustainable Energy Rev* 2016;60:1634–53.
- [44] Edmunds RK, Cockerill TT, Foxon TJ, Ingham DB, Pourkashanian M. Technical benefits of energy storage and electricity interconnections in future British power systems. *Energy* 2014;70:577–87. <https://doi.org/10.1016/j.energy.2014.04.041>.
- [45] Yang L, Yan H, Lam JC. Thermal comfort and building energy consumption implications - a review. *Appl Energy* 2014;115:164–73. <https://doi.org/10.1016/j.apenergy.2013.10.062>.

- [46] Pillai JR, Heussen K, Østergaard PA. Comparative analysis of hourly and dynamic power balancing models for validating future energy scenarios. *Energy* 2011;36(5): 3233–43. <https://doi.org/10.1016/j.energy.2011.03.014>.
- [47] Kwon PS, Østergaard PA. Priority order in using biomass resources - Energy systems analyses of future scenarios for Denmark. *Energy* 2013;63:86–94. <https://doi.org/10.1016/j.energy.2013.10.005>.
- [48] Duquette J, Wild P, Rowe A. The potential benefits of widespread combined heat and power based district energy networks in the province of Ontario. *Energy* 2014; 67:41–51. <https://doi.org/10.1016/j.energy.2013.12.038>.
- [49] Østergaard PA, Lund H, Mathiesen BV. Energy system impacts of desalination in Jordan. *Int J Sustainable Energy Plann Manag* 2014;1:29–40. <https://doi.org/10.5278/ijsepm.2014.1.3>.
- [50] Østergaard PA. Regulation strategies of cogeneration of heat and power (CHP) plants and electricity transit in Denmark. *Energy* 2010;35(5):2194–202. <https://doi.org/10.1016/j.energy.2010.02.005>.
- [51] Onisanwa ID, Adaji MO. Electricity consumption and its determinants in Nigeria. *J Econ Manag* 2020;41(3):87–104.
- [52] Tagliapietra S, Occhiali G, Nano E, Kalcik R. The impact of electrification on labour market outcomes in Nigeria. no. 3. Springer International Publishing; 2020.
- [53] Ibrahim H, Kirkil G. Electricity demand and supply scenario analysis for Nigeria using long range energy alternatives planning (LEAP). *J Sci Res Rep* 2018;19(2): 1–12. <https://doi.org/10.9734/jsrr/2018/39719>.
- [54] Ezennaya SO, Isaac O, Ezeanyim OC. Analysis of Nigeria's national electricity demand forecast (2013–2030). *Int J Sci Technol Res* 2014;3(3):333–40.
- [55] Bamisile O, Huang Q, Xu X, Hu W, Liu W, Liu Z, et al. An approach for sustainable energy planning towards 100 % electrification of Nigeria by 2030. *Energy* 2020; 197:117172. <https://doi.org/10.1016/j.energy.2020.117172>.
- [56] Ezennaya SO, Isaac OE, Okolie UO, Ezeanyim OC. Analysis Of Nigeria's national electricity demand forecast (2013–2030). *Int J Sci Technol Res* 2014;3(3):333–40.
- [57] Commission NER. Power generation in Nigeria.
- [58] Oji JO, Idusuyi N, Aliu TO, Petinrin MO, Odejebi OA, Adetunji AR. Utilization of solar energy for power generation in Nigeria. *Int J Energy Eng* 2012;2(2):54–9. <https://doi.org/10.5923/j.ijee.20120202.07>.
- [59] Ogunmodimu O, Okoroigwe EC. Concentrating solar power technologies for solar thermal grid electricity in Nigeria: a review. *Renewable Sustainable Energy Rev* 2018;90(February):104–19. <https://doi.org/10.1016/j.rser.2018.03.029>.
- [60] Giwa A, Alabi A, Yusuf A, Olukan T. A comprehensive review on biomass and solar energy for sustainable energy generation in Nigeria. *Renewable Sustainable Energy Rev Mar.* 2017;69:620–41. <https://doi.org/10.1016/J.RSER.2016.11.160>.
- [61] Brimmo AT, Sodiq A, Sofela S, Kolo I. Sustainable energy development in Nigeria: Wind, hydropower, geothermal and nuclear (Vol. 1). In: *Renew. Sustain. Energy Rev.*; 2017. p. 474–90. <https://doi.org/10.1016/j.rser.2016.11.162>.
- [62] Ajayi OO, Ohijeagbon OD, Ajanaku KO, Aasa SA, Omotosho OA. Techno-economic assessment of renewable electricity for rural electrification and IT applications in selected sites across the geopolitical zones of Nigeria. 23rd International Business Information Management Association Conference, IBIMA 2014. 2014.
- [63] Adekoya LO, Adewale AA. Wind energy potential of Nigeria. *Renew. Energy* 1992; 2(1):35–9. [https://doi.org/10.1016/0960-1481\(92\)90057-A](https://doi.org/10.1016/0960-1481(92)90057-A).
- [64] Olanipekun BA, Adelakun NO. Assessment of renewable energy in Nigeria: Challenges and benefits. *SSRG Int J Eng Trends Technol* 2020;68(1):64–7. <https://doi.org/10.14445/22315381/IJETT-V68I1P209>.
- [65] N. E. R. Commission, Power Generation in Nigeria, [Online]. Available: <<http://www.nercng.org/index.php/home/nesi/403-generation>>; 2019 [accessed: 04-Apr-2019].
- [66] Rehman S, Al-Hadhrami LM, Alam MM. Pumped hydro energy storage system: a technological review. *Renewable Sustainable Energy Rev* 2015;44:586–98. <https://doi.org/10.1016/j.rser.2014.12.040>.
- [67] EnergyPLAN, Advanced energy system analysis model.
- [68] E. M. Team and S. Energy, EnergyPLAN, pp. 0–10.
- [69] Connolly D. EnergyPLAN Cost Database: Version 4, no. January, pp. 0–15, 2018.
- [70] Lund H, Sorknæs P, Mathiesen BV, Hansen K. Beyond sensitivity analysis: a methodology to handle fuel and electricity prices when designing energy scenarios. *Energy Res Soc Sci* 2018;39:108–16. <https://doi.org/10.1016/j.erss.2017.11.013>.
- [71] Mathiesen SR, Vad Brian, Lund Henrik, Hansen Kenneth, Ridjan Iva, Djørup R, et al. IDA's Energy Vision 2050 A Smart Energy System strategy for 100% renewable Denmark. In: *Dep. Dev. Planning, Aalborg Univ.*; 2015. p. 1–156.
- [72] Bamisile O, Li J, Huang Q, Obiora S, Ayambire P, Zhang Z, et al. Environmental impact of hydrogen production from Southwest China's hydro power water abandonment control. *Int J Hydrogen Energy* 2020;45(46):25587–98. <https://doi.org/10.1016/j.ijhydene.2020.06.289>.
- [73] Glenk G, Reichelstein S. Economics of converting renewable power to hydrogen. *Nat Energy* 2019;4(3):216–22. <https://doi.org/10.1038/s41560-019-0326-1>.
- [74] Davis SJ, Lewis NS, Shaner M, Aggarwal S, Arent D, Azevedo IL, et al. Net-zero emissions energy systems. *Science* 2018;360(6396):eaas9793. <https://doi.org/10.1126/science.aas9793>.
- [75] Safari F, Dincer I. Development and analysis of a novel biomass-based integrated system for multigeneration with hydrogen production. *Int J Hydrogen Energy* 2019;44(7):3511–26. <https://doi.org/10.1016/j.ijhydene.2018.12.101>.
- [76] Li D, Zouma A, Liao JT, Yang HT. An energy management strategy with renewable energy and energy storage system for a large electric vehicle charging station. p. 100076. In: *eTransportation*; 2020. <https://doi.org/10.1016/j.etrans.2020.100076>.
- [77] IEA, Global EV Outlook 2019; scaling up the transition to electric mobility; 2019.
- [78] Hardman S, Jenn A, Tal G, Axsen J, Beard G, Daina N, et al. A review of consumer preferences of and interactions with electric vehicle charging infrastructure. *Transp Res Part D Transp Environ* 2018;62:508–23. <https://doi.org/10.1016/j.trd.2018.04.002>.
- [79] Levinson RS, West TH. Impact of public electric vehicle charging infrastructure. *Transp Res Part D Transp Environ* 2018;64:158–77. <https://doi.org/10.1016/j.trd.2017.10.006>.
- [80] Connolly D, Lund H, Mathiesen BV, Leahy M. The first step towards a 100% renewable energy-system for Ireland. *Appl Energy* 2011;88(2):502–7. <https://doi.org/10.1016/j.apenergy.2010.03.006>.
- [81] World Energy Council, World Energy Trilemma Index, p. 156, 2018, <https://doi.org/10.1089/jpm.2004.7.865>.
- [82] Oyewo AS, Aghahosseini A, Christian B. Assessment of energy storage technologies in transition to a 100 % renewable energy system for Nigeria. 11th International Renewable Energy Storage Conference. 2017. no. March.